Space Exploration & Aerospace Engineering

Part III

Satellites in Earth Orbit

About sixteen thousand active satellites now circle the Earth. This Part explains where they fly (LEO, MEO, GEO and special orbits), how constellations are designed, how a satellite bus is built, and the families of payload instruments they carry.



Chapters 11–12 · pages 181–214 of the complete manual
First Edition · Current to October 2026 · Written in plain language with deep technical detail

ENGINEERING REFERENCE MANUAL

Space Exploration & Aerospace Engineering

A complete, illustrated guide to the space ecosystem — from stratospheric balloons to interstellar probes — explaining how every major class of space system works, how it is engineered, and where artificial intelligence is changing it.

First Edition · Information current to October 2026
A4 format · Fourteen Parts · Fifty-three chapters · Ten appendices
Version 1.0 · October 2026

About this edition

Space Exploration & Aerospace Engineering — An Engineering Reference Manual. First Edition, Version 1.0, October 2026.

This manual was written to be read by students, engineers, analysts, managers and curious readers. Each topic starts with plain-language explanations and builds up to the equations, numbers and design trade-offs that practising engineers use. Sentences are kept short and every technical term is explained in plain words when it first appears; the glossary (Appendix H) and the list of acronyms (Appendix G) collect them. Facts, mission status and programme milestones are current to October 2026; space programmes change quickly, so dates of future events should be read as plans, not certainties.

Figures, photographs and data

Diagrams, charts and graphs were drawn specifically for this manual from published physical relationships and public data. Photographs are from the NASA Image and Video Library and are used under NASA's media usage guidelines (NASA material is generally not subject to copyright in the United States); individual credits appear with each image and in Appendix J. NASA does not endorse this publication. Numerical values (masses, power levels, performance) are representative published figures and may differ slightly between sources and mission phases.

Safety and responsibility

This is an educational reference. It is not a design standard, a flight rule, or a substitute for the requirements, analyses and verification processes of a qualified engineering organisation. Chapters dealing with military and intelligence systems use only openly published information and describe principles, not operational or classified details.

How numbers are written

SI units are used throughout, with common aerospace units (nautical miles, feet, pounds-force) shown where the industry uses them. Large numbers use thin spacing (e.g., 35 786 km). "≈" means approximately, "~" means order-of-magnitude. Dates follow the ISO format (YYYY-MM-DD) in tables.

Typography and production

Set in Source Serif 4, Inter and JetBrains Mono. Equations typeset with MathJax. Produced from Markdown sources with an automated pipeline (pandoc, Chromium, pypdf); the sources and build scripts accompany the PDF.

Contents of Part III

PART III · SATELLITES IN EARTH ORBIT

11 Earth Orbits, Constellations and Small Satellites 182

11.1 Choosing an Orbit 182
11.2 Coverage Geometry 184
11.3 Constellation Design 186
11.4 The Population of Earth Orbit 189
11.5 Mega-Constellations 189
11.6 The Geostationary Orbit 192
11.7 Small Satellites and the CubeSat Revolution 192
11.8 Launch and Early Operations 194
11.9 Orbit Maintenance 195
11.10 A Worked Small-Satellite Design 195
11.11 Getting Small Satellites to Orbit 197
11.12 Formation Flying and Swarms 197
11.13 Operating Large Constellations 198
11.14 Sustainability, Regulation and the Night Sky 198

12 Satellite Buses and Payload Instruments 201

12.1 The Bus as a Product 201
12.2 A Taxonomy of Payload Instruments 203
12.3 Optical and Infrared Instruments 205
12.4 Microwave Instruments 207
12.5 Lidar Instruments 208
12.6 In-Situ Instruments: Fields and Particles 208
12.7 Service Payloads 209
12.8 Performance, Calibration and Data 209
12.9 Designing an Optical Payload: Radiometry and SNR 211
12.10 Instrument Case Studies 212
12.11 Large Communications Platforms: Power as the Measure 212

This volume is one Part of *Space Exploration & Aerospace Engineering — An Engineering Reference Manual*. Page numbers, chapter numbers and figure numbers are the same as in the complete manual, so cross-references to other chapters (for example “Chapter 24”) and to the appendices refer to the complete book.



PART III

Satellites in Earth Orbit

About sixteen thousand active satellites now circle the Earth. This Part explains where they fly (LEO, MEO, GEO and special orbits), how constellations are designed, how a satellite bus is built, and the families of payload instruments they carry.

11 Earth Orbits, Constellations and Small Satellites 12 Satellite Buses and Payload Instruments

Image: NASA/KSC (NASA/Randy Beaudoin)

CHAPTER

11 Earth Orbits, Constellations and Small Satellites



Image: NASA/JSC (Peggy Whitson)

A satellite's orbit decides what it can see, how often, with what delay and at what cost. This chapter turns the orbital mechanics of Chapter 3 into practical mission design - choosing an orbit, calculating coverage and revisit, and designing constellations of many satellites. It then surveys the crowded reality of Earth orbit in 2026, the mega-constellations that dominate it, the geostationary belt, the small-satellite and CubeSat revolution, rideshare and in-orbit deployment, formation flying, and the sustainability questions raised by tens of thousands of satellites.

IN THIS CHAPTER

11.1 Choosing an Orbit
11.2 Coverage Geometry
11.3 Constellation Design
11.4 The Population of Earth Orbit
11.5 Mega-Constellations
11.6 The Geostationary Orbit
11.7 Small Satellites and the CubeSat Revolution
11.8 Launch and Early Operations
11.9 Orbit Maintenance
11.10 A Worked Small-Satellite Design
11.11 Getting Small Satellites to Orbit
11.12 Formation Flying and Swarms
11.13 Operating Large Constellations
11.14 Sustainability, Regulation and the Night Sky

11.1 Choosing an Orbit

Orbit selection is usually the first and most important decision in mission design, because it sets the environment (Chapter 2), the launch cost, the communication geometry and the performance of the payload. The main drivers are:

- **What must be seen or served** — the whole globe, high latitudes, a specific region, or a point in deep space.
- **How often** — continuously (communications, early warning), several times a day (weather, maritime), daily (crop monitoring), or occasionally (mapping).
- **How closely** — resolution improves in proportion to distance for optical and radar sensors, and link budgets improve with the square of distance.
- **Lighting** — optical imagers want consistent Sun angles; thermal and power design prefer predictable eclipses.
- **Latency** — interactive services (voice, gaming, finance) want short round-trip delays.
- **Environment and lifetime** — drag, radiation, debris density and disposal rules.
- **Launch and cost** — higher-energy orbits need bigger rockets or longer transfers.

Table 11.1 gives typical choices.

TABLE 11.1 Typical orbit choices by mission type.

Mission	Typical orbit	Reason
High-resolution optical imaging	400–700 km sun-synchronous (10:00–10:30 LTDN), increasingly VLEO 250–400 km	Short range for resolution; constant lighting
Radar imaging (SAR)	500–700 km dawn–dusk sun-synchronous	Near-continuous sunlight for power; radar works day/night
Global weather (polar)	800–850 km sun-synchronous, morning and afternoon planes	Full global coverage twice daily
Weather (regional, continuous)	GEO	Images every 30 s–10 min over a hemisphere
Ocean altimetry	1 336 km, 66° (Jason, Sentinel-6); 890 km, 77.6° (SWOT)	Avoid sun-synchronous tidal aliasing; long-lived repeat tracks
Gravity mapping	450–500 km polar, low-low formation (GRACE-FO)	Sensitivity to gravity anomalies
Broadband internet	340–1 200 km LEO (constellations); GEO; MEO (O3b)	Low latency (LEO) or wide coverage per satellite (GEO)
Direct-to-device mobile	500–700 km LEO; GEO for narrowband	Short range to tiny phone antennas
Navigation	19 000–23 300 km MEO, GEO/IGSO for regional	Many satellites visible at once; stable orbits
Missile warning	GEO and HEO (Molniya-type), increasingly LEO/MEO layers	Continuous view of launch regions; polar coverage
Space science (astronomy)	LEO (Hubble), HEO (Chandra), Sun–Earth L2 (JWST, Euclid, Roman)	Cold, stable, unobstructed sky
Solar wind monitoring	Sun–Earth L1	Upstream of Earth, ≈ 1 hour warning
Technology demos, education	400–600 km (rideshare or ISS deployment)	Cheap launch, natural decay

11.2 Coverage Geometry

11.2.1 Footprint and Elevation

What a satellite can see from altitude h is described by four angles (Figure 11.1). The **elevation angle** ε is the angle of the satellite above the local horizon of a ground point. The **nadir angle** η is the angle at the satellite between the nadir and the ground point; the others are the **Earth central angle** λ and the slant range ρ. They are linked by

$$\sin\eta = \frac{R_\oplus}{R_\oplus + h}\cos\varepsilon, \qquad \lambda = 90^\circ - \eta - \varepsilon, \qquad \rho = R_\oplus \frac{\sin\lambda}{\sin\eta}$$

and the area of the coverage circle is $A = 2\pi R_\oplus^2(1-\cos\lambda)$.

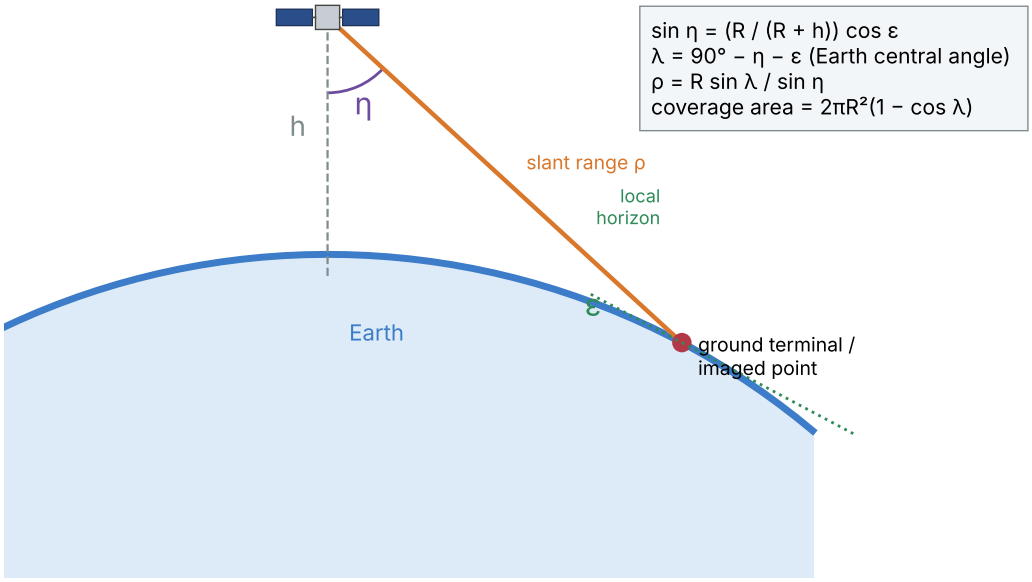


FIGURE 11.1 Coverage geometry of a satellite. The minimum elevation angle at which a terminal can use the satellite — limited by terrain, buildings, atmosphere and antenna design — sets the coverage radius.

The **minimum elevation angle** is a key design parameter. It is 5–10° for ground stations with clear horizons. For broadband user terminals it is 25–40° (to avoid blockage and long atmospheric paths and to share spectrum with GEO systems); for navigation it is 10–20°. A higher minimum elevation shrinks the footprint, so more satellites are needed (Figure 11.2).

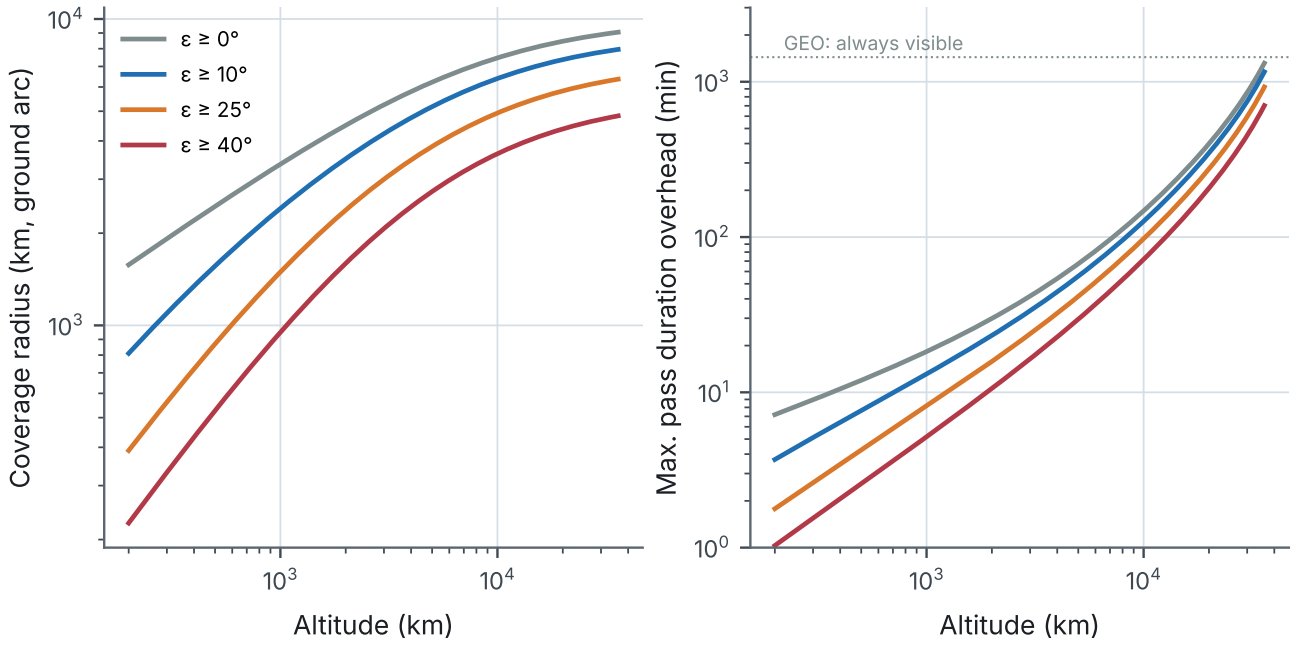


FIGURE 11.2 Left: coverage radius on the ground versus altitude for four minimum elevation angles. Right: maximum duration of an overhead pass. LEO satellites are visible for minutes, MEO for hours, GEO continuously.

WORKED EXAMPLE — HOW MANY MINUTES CAN A GROUND STATION TALK TO A 550 KM SATELLITE?

For h = 550 km and ε = 10°: $\sin\eta = (6\,378/6\,928)\cos 10^\circ = 0.9066$, so $\eta = 65.0^\circ$ and $\lambda = 90 - 65.0 - 10 = 15.0^\circ$. The coverage radius is $6\,378 \times 0.262 = 1\,670$ km. The satellite's angular rate is its mean motion, $n = 360^\circ/95.6$ min = 3.77°/min, so an overhead pass lasts at most $2 \times 15.0/3.77 \approx 8.0$ minutes; passes that do not go directly overhead are shorter. A station sees such a satellite on 4–6 passes a day at mid-latitudes, but on almost every orbit from Svalbard if the orbit is polar.

11.2.2 Swath, Revisit and Repeat

For an Earth-observation satellite, the relevant footprint is the **swath** — the width of ground imaged as the satellite moves. It is set by the instrument field of view or, for agile satellites, by how far off-nadir it can point. Two different time measures are used:

- **Revisit time** — how often a given place *can* be observed (possibly at different angles). Agile pointing and constellations reduce revisit to hours or less.
- **Repeat cycle** — how long until the satellite passes over exactly the same ground track (Chapter 3), giving identical viewing geometry. Landsat 8 and 9 each repeat every 16 days; together they give 8-day coverage.

For a single sun-synchronous imager with swath W at the equator, global coverage takes about $2\pi R_\oplus/(W \times N_{orbits/day})$ days. For example, a 290 km swath (Sentinel-2) with 14.3 orbits per day covers the equator in about 10 days. Two satellites halve this to 5 days. Revisit improves at higher latitudes, where ground tracks converge.

11.3 Constellation Design

When one satellite is not enough — for continuous coverage, frequent revisit, more capacity or redundancy — engineers design **constellations**.

11.3.1 Walker Constellations

The most widely used patterns were described by J. G. Walker in 1970–1984. A **Walker delta** constellation is written **i: T/P/F** — inclination, total number of satellites T, number of equally spaced orbital planes P (with RAAN spaced 360°/P), and a phasing parameter F (0 to P−1) that sets the relative position of satellites in adjacent planes (in units of 360°/T). GPS's original design is close to a 55°: 24/6/1 Walker; Galileo is a 56°: 24/3/1 Walker (plus spares). A **Walker star** (or polar, "street-of-coverage") constellation places planes over 180° of RAAN with near-polar inclination, so that adjacent planes overlap like streets; Iridium uses 66 active satellites in six planes at 86.4° (Figure 11.3).

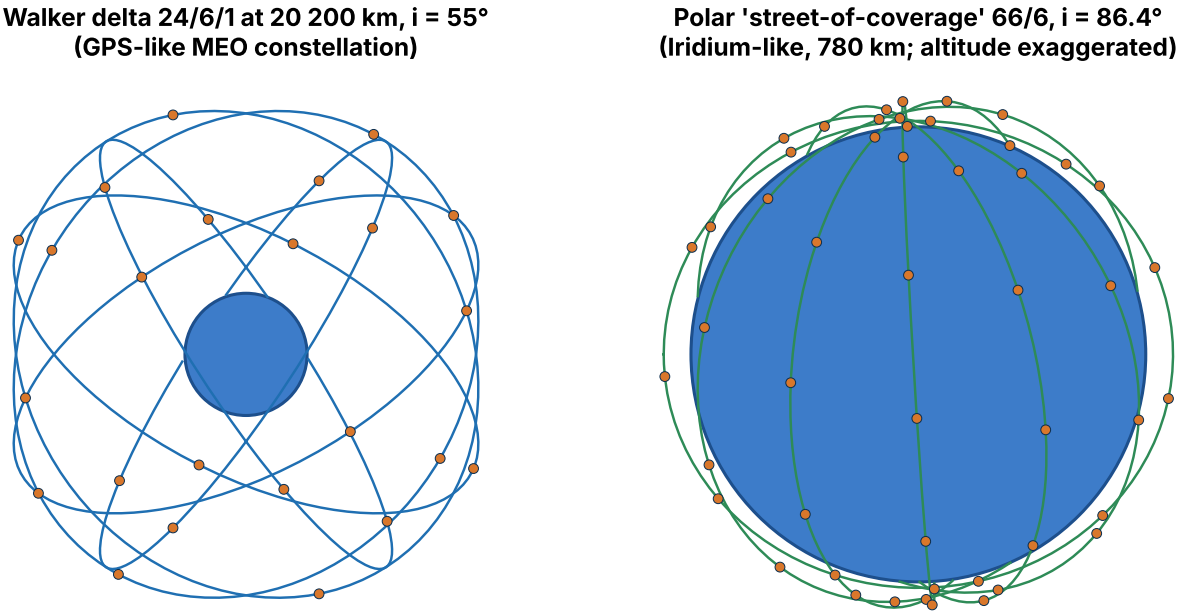


FIGURE 11.3 Two classic constellation geometries: a Walker delta MEO constellation like GPS, and a near-polar "street-of-coverage" constellation like Iridium.

Other families include **Flower constellations** (satellites on identical repeating ground tracks) and **elliptical constellations** that concentrate coverage over high latitudes (Molniya, Ellipso). There are also hybrid multi-orbit systems combining LEO, MEO and GEO (e.g., Eutelsat OneWeb with Eutelsat's GEO fleet; SES with O3b mPOWER and GEO; IRIS^2).

11.3.2 How Many Satellites?

The number of satellites needed for continuous global coverage falls rapidly with altitude because each satellite sees more of the Earth (Figure 11.4). Three GEO satellites cover everything except the polar caps; about 24 MEO satellites give global navigation (which needs at least four in view everywhere); 66 satellites at 780 km give Iridium global voice coverage. Broadband mega-

constellations use far more satellites than coverage alone requires — thousands. This is because what limits them is **capacity**. Each satellite can serve only so many users with its spectrum and beams, and users want high elevation angles and low latency.

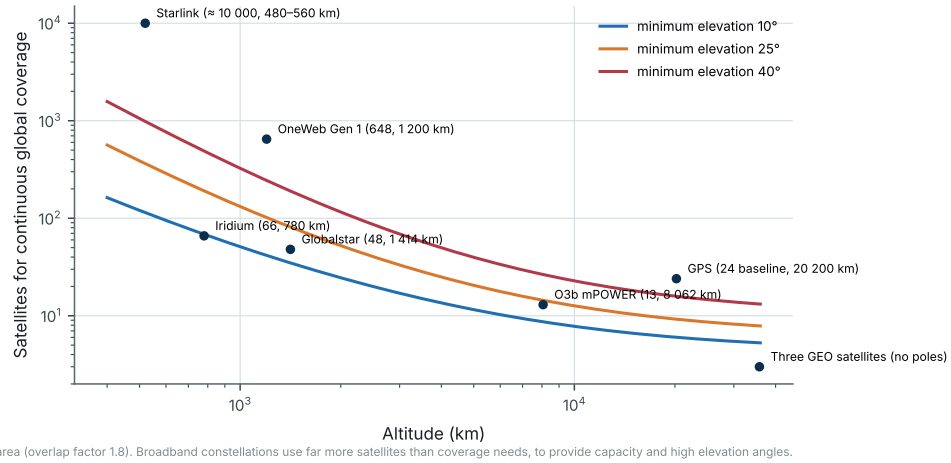


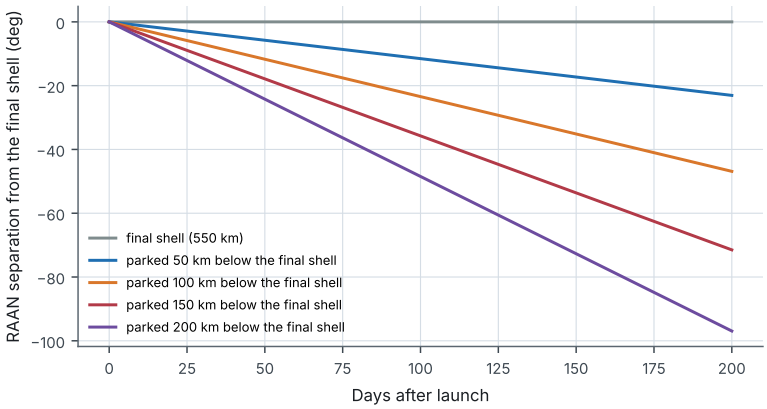
FIGURE 11.4 Approximate number of satellites needed for continuous single global coverage versus altitude, compared with real systems. Broadband constellations exceed the coverage requirement many times over to provide capacity.

The main constellation trade-offs are:

Choice	Lower altitude	Higher altitude
Number of satellites	More	Fewer
Latency (round trip, user–satellite–gateway)	≈ 20–40 ms in LEO	≈ 125 ms in MEO, ≈ 500+ ms in GEO
Link budget, user terminal size	Better / smaller	Worse / larger
Satellite lifetime vs drag	Shorter (natural disposal)	Longer (must actively dispose)
Radiation	Lower	Higher (especially MEO)
Handover and tracking	Frequent (minutes)	Rare or none
Launch cost per satellite	Lower	Higher
Debris risk if satellites fail	Self-cleaning below ≈ 550 km	Persistent above ≈ 600 km

11.3.3 Deploying a Constellation: Plane Phasing

Placing satellites in many orbital planes is expensive if each plane needs its own launch. Constellation operators therefore exploit **differential nodal precession** (Chapter 3). Because the J_2 precession rate depends on altitude, satellites from a single launch can wait at different "parking" altitudes, where their planes drift apart in RAAN at different rates. They then raise themselves to the operational shell at the right moment (Figure 11.5). A satellite parked 100 km below a 550 km shell drifts about 0.18° per day relative to it, so three months of waiting separates planes by about 15°. The cost is time rather than propellant — a trade that suits electric propulsion and mass-produced satellites.



J_2 nodal drift at 53° inclination. Satellites from one launch wait at different altitudes and drift apart in RAAN for free before climbing to the operational shell — 'differential precession' plane phasing.

FIGURE 11.5 Relative RAAN drift between satellites parked below a 550 km, 53° shell and satellites already in it. Waiting at lower altitudes spreads orbital planes using Earth's oblateness instead of propellant.

TABLE 11.2 Key parameters of major satellite constellations (October 2026).

Constellation	Operator	Service	Altitude / inclination	Satellites (in orbit / planned)	Mass per satellite	Notable features
Starlink	SpaceX (USA)	Broadband, D2C, government	≈ 480–570 km; 33°–97.6°	≈ 10 650 / 12 000 (Gen1) + Gen2	≈ 300 kg (V1.5), ≈ 800 kg (V2 Mini), ≈ 2 t (V3)	Laser mesh, argon Hall thrusters, Starship-launched V3
Eutelsat OneWeb	Eutelsat (France/UK)	Enterprise broadband	1 200 km, 87.9°	≈ 650 / 648 + Gen2	≈ 150 kg	Bent-pipe Ku/Ka; GEO + LEO offering
Amazon Leo	Amazon (USA)	Broadband	590–630 km, 33°–52°	≈ 400 / 3 232	≈ 600 kg	Launches on five vehicle types
Qianfan	SSST (China)	Broadband	≈ 800–1 000 km (first shells)	≈ 220–240 / > 14 000	≈ 250–300 kg	Launched in batches of 18 on Long March 6A
Guowang	China SatNet	Broadband, government	≈ 500–1 145 km	≈ 200 / ≈ 13 000	various	State-owned
Iridium NEXT	Iridium (USA)	Voice, data, IoT, PNT	780 km, 86.4°	66 active + spares	860 kg	Cross-links; hosted ADS-B (Aireon)
Globalstar	Globalstar (USA)	Voice, data, Apple emergency SOS	1 414 km, 52°	≈ 25 active + new generation	≈ 700 kg	Bent-pipe; new satellites from MDA
O3b mPOWER	SES (Luxembourg)	High-throughput MEO	8 062 km, 0°	13	≈ 1 700 kg	Thousands of steerable beams
AST SpaceMobile	AST (USA)	Direct-to-device	≈ 520–700 km	13 / ≈ 45 (2027)	≈ 1.5–6 t	Giant phased arrays
SDA PWSA	US Space Development Agency	Military data transport, missile tracking	≈ 1 000 km	42 / ≈ 1 000 (all tranches)	≈ 500–1 000 kg	Optical crosslinks, Link 16
Planet (Dove/Pelican)	Planet Labs (USA)	Earth observation	≈ 475–525 km SSO	≈ 200	5 kg (Dove), ≈ 100–200 kg (Pelican)	Daily global imaging

Constellation	Operator	Service	Altitude / inclination	Satellites (in orbit / planned)	Mass per satellite	Notable features
Spire	Spire Global (USA)	RF data (AIS, ADS-B, GNSS-RO)	≈ 400–600 km	≈ 80–100	≈ 5 kg (3U)	GNSS radio occultation for weather
ICEYE	ICEYE (Finland)	SAR imaging	≈ 550 km SSO	≈ 50	≈ 90–130 kg	Largest SAR constellation

11.4 The Population of Earth Orbit

The orbital population in 2026 is dominated by a few altitude bands (Figure 11.6). CelesTrak's public catalogue lists about **35 000 objects** in Earth orbit — 20 100 payloads (of which about 17 200 are operational), 12 500 debris fragments and 2 300 rocket bodies. More than half of all objects now lie between 450 and 600 km, where Starlink operates. A second, older concentration between 750 and 1 000 km contains sun-synchronous Earth-observation satellites and Iridium. It also holds the long-lived debris from the 2007 Fengyun-1C test and the 2009 Iridium–Cosmos collision.

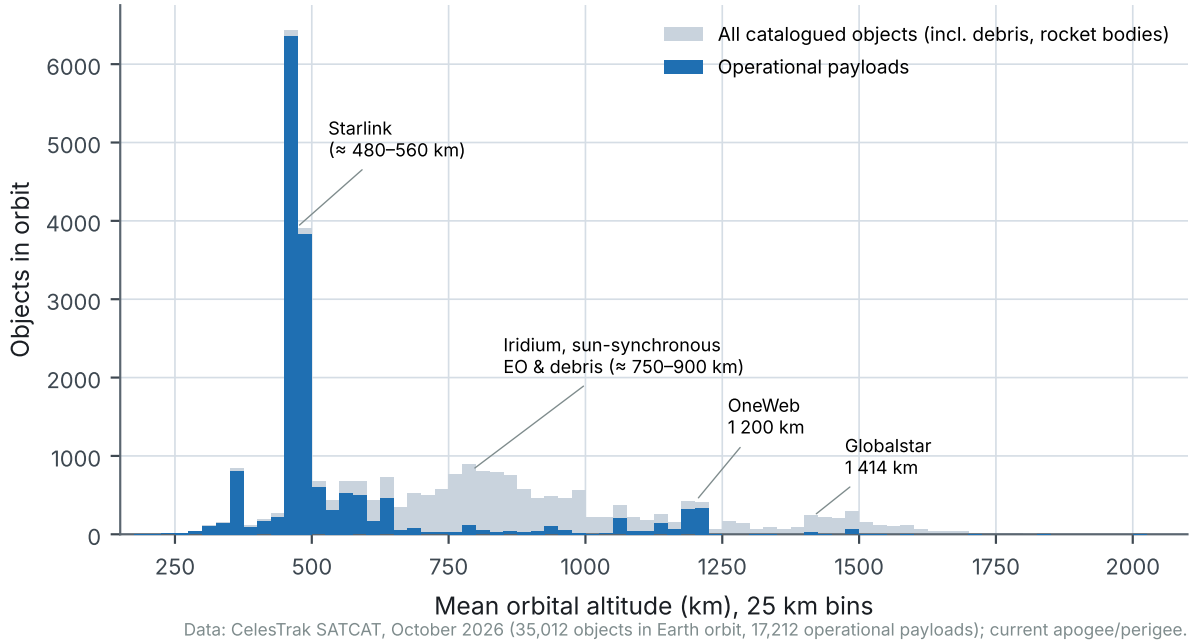


FIGURE 11.6 Distribution of catalogued objects in Earth orbit by mean altitude, October 2026. Operational payloads (blue) cluster in constellation shells; debris (grey) dominates the 700–1 000 km region.

11.5 Mega-Constellations

11.5.1 Starlink

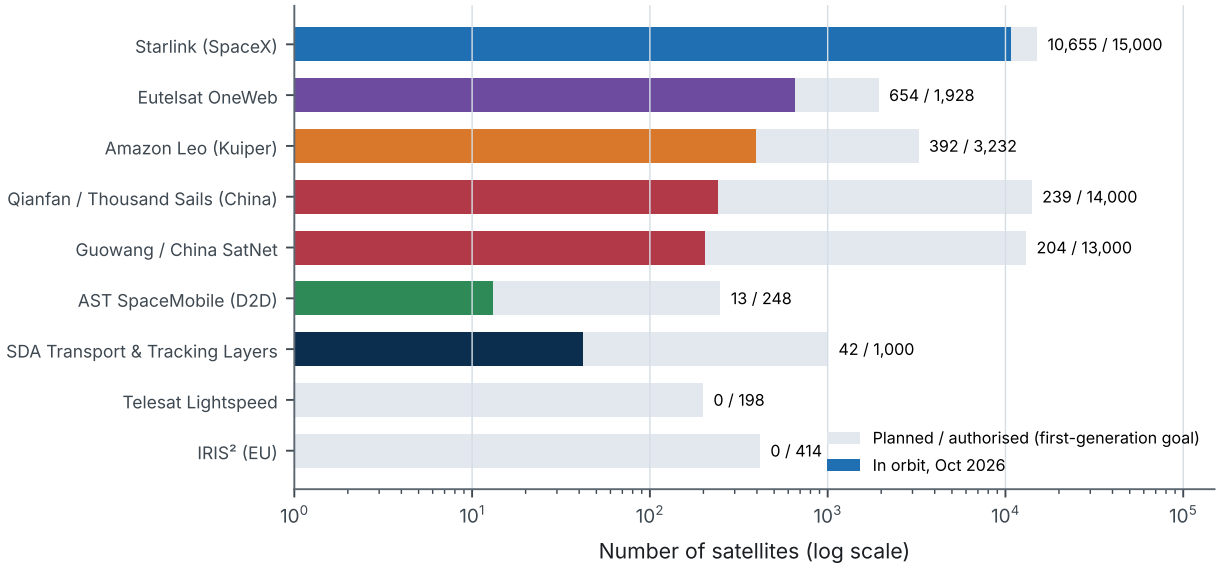
SpaceX's **Starlink** is the largest satellite system ever built. Its first operational satellites launched in May 2019. By October 2026 more than 10 600 were in orbit, serving more than 8 million subscribers in over 140 countries and territories. They provide broadband, maritime, aviation and mobile ("Direct to Cell") services. Key engineering features include:

- **Mass production** — satellites are built at rates of several per day in Redmond, Washington. They are launched in batches of 21–29 (V2 Mini) on Falcon 9 or, from September 2026, in larger batches of the much bigger **V3** satellites on Starship.
- **Flat-pack design** — satellites stack in the fairing without a dispenser; a single solar array per satellite on V1/V1.5, two on V2 Mini.
- **Electric propulsion** — krypton (V1) and argon (V2 Mini) Hall thrusters for orbit raising, station-keeping, collision avoidance and de-orbit.
- **Phased-array antennas** for Ku- and Ka-band user and gateway links and E-band backhaul; **laser inter-satellite links** (three or more per satellite) forming a mesh network that routes traffic across oceans and polar regions.
- **Autonomous collision avoidance** — satellites manoeuvre automatically using conjunction data; SpaceX reports tens of thousands of avoidance manoeuvres per half-year (Chapter 21).
- **Low orbits** for natural disposal: SpaceX began lowering about 4 400 satellites from about 550 km to about 480 km during 2026, citing lower debris density and much faster decay of failed satellites.

The US government uses a derivative, **Starshield**, for national-security applications, including a proliferated reconnaissance constellation for the National Reconnaissance Office.

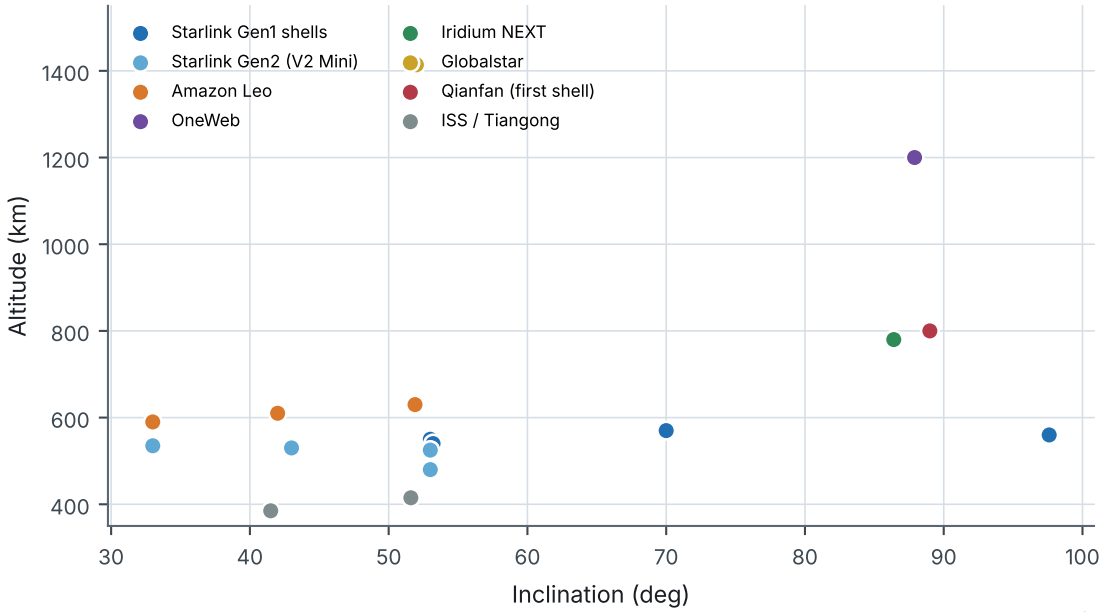
11.5.2 Other Broadband Constellations

- **Eutelsat OneWeb** — about 650 satellites at 1 200 km in 12 near-polar planes, serving enterprise, government, aviation and maritime customers through partners; second-generation satellites are being procured.
- **Amazon Leo** (formerly Project Kuiper) — authorised for 3 232 satellites in three shells at 590–630 km. About 400 were in orbit by mid-2026 after launches on Atlas V, Falcon 9, Vulcan, Ariane 6 and New Glenn. Beta service was planned for autumn 2026. The FCC waived its July 2026 half-constellation milestone.
- **Qianfan / "Thousand Sails"** (Shanghai Spacecom Satellite Technology) — planned to exceed 14 000 satellites; about 220–240 launched by mid/late 2026.
- **Guowang** (China Satellite Network Group, "SatNet") — a state constellation of about 13 000 satellites; about 200 launched by September 2026.
- **Telesat Lightspeed** — 198 advanced satellites with optical links planned for enterprise and government service from the late 2020s.
- **IRIS²** — the EU's secure multi-orbit constellation (SpaceRISE consortium of SES, Eutelsat and Hispasat). Under an August 2026 agreement, it comprises 330 LEO and 18 MEO satellites plus 66 dedicated government satellites, with first launches planned for 2029.
- **AST SpaceMobile** — very large (up to ≈ 220 m^2 phased-array) "BlueBird" satellites that connect directly to unmodified phones; 13 in orbit by August 2026, with about 45 targeted by early 2027 (Chapter 17).



In-orbit counts: GCAT (Oct 2026) and company/agency reports; planned numbers are first-generation authorisations or declared targets and change frequently.

FIGURE 11.7 Mega-constellations: satellites in orbit (coloured) compared with first-generation plans or authorisations (grey), October 2026.



Representative published shells; constellations are reconfigured over time (e.g., Starlink lowering shells from ≈ 550 to ≈ 480 km during 2026).

FIGURE 11.8 Altitudes and inclinations of representative constellation shells. Shells at different inclinations concentrate capacity where most users live (mid-latitudes) while polar shells cover high latitudes.

11.5.3 Military Proliferated Constellations

The US **Space Development Agency (SDA)** is deploying the **Proliferated Warfighter Space Architecture** — hundreds of relatively low-cost satellites in LEO in successive "tranches" every two years. It has a **Transport Layer** of data-relay satellites with optical crosslinks and Link 16 tactical radios. It also has a **Tracking Layer** of infrared satellites that detect and track missiles (Chapter 20). Tranche 1 launches began in September 2025. China, Europe and others are pursuing similar concepts, reflecting a shift from a few exquisite satellites to resilient proliferated architectures.

11.6 The Geostationary Orbit

11.6.1 A Unique Resource

A satellite at 35 786 km altitude above the equator, moving eastward with a period of one sidereal day, appears fixed in the sky. Antennas on the ground need not track it, and three satellites cover most of the populated Earth. The **geostationary arc** is therefore a limited natural resource, shared through ITU coordination. Satellites are assigned **orbital slots** (longitudes), typically spaced 2–3° for the same frequency bands. They must stay within a **station-keeping box** of about ±0.05–0.1° in longitude and latitude. About 1 340 payloads are in GEO-type orbits, of which roughly 600–650 are operational.

Natural forces push GEO satellites out of their boxes. The Sun and Moon tilt the orbit by about 0.8° per year (north–south drift). The ellipticity of Earth's equator causes east–west drift towards stable points near 75° E and 105° W, and solar radiation pressure changes the eccentricity. Station-keeping requires about 50 m/s per year (Chapter 3). At end of life, operators must raise their satellites at least 235 km plus a margin depending on area-to-mass ratio above GEO into a **graveyard orbit** (IADC guideline). Some operators extend life by allowing inclination to grow ("inclined-orbit operation") or by using life-extension servicers (Chapter 40).

11.6.2 Eclipses and Sun Interference

GEO satellites experience eclipses only around the equinoxes (up to 72 minutes per day for about 45 days in each season). Around the equinoxes, the Sun also passes directly behind satellites as seen from ground antennas, causing **Sun outages** of a few minutes per day due to solar noise.

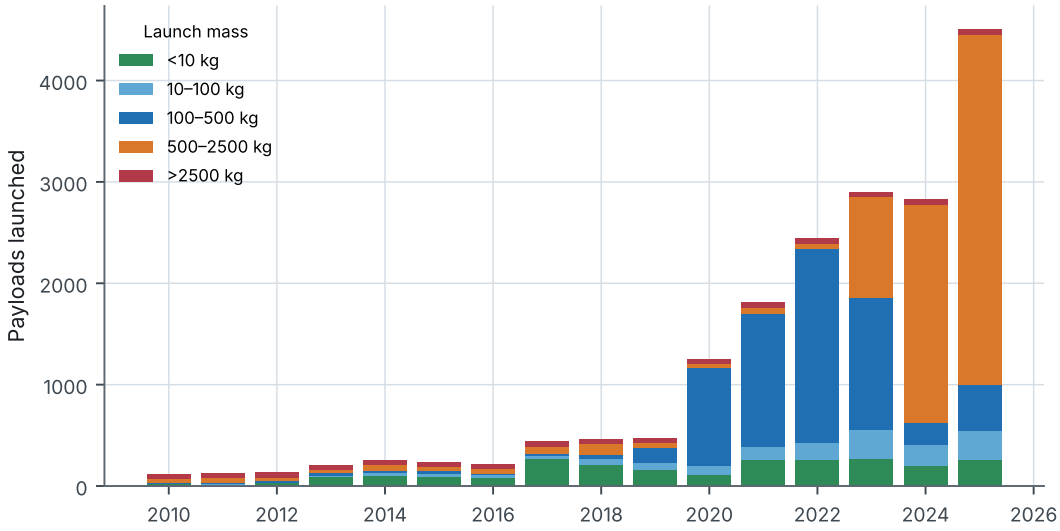
11.6.3 Trends in GEO

GEO remains the workhorse for broadcasting and many government services but faces competition from LEO broadband. Trends include **software-defined, reconfigurable payloads** that can change coverage and frequency plans in orbit (Airbus OneSat, Thales Inspire, Boeing 702X). Other trends are **all-electric propulsion**, **smaller GEO satellites** (Astranis MicroGEO, about 400 kg), and **in-orbit servicing** to extend life.

11.7 Small Satellites and the CubeSat Revolution

11.7.1 Why Small?

"Small satellites" (below about 500 kg) have transformed the industry by trading individual capability for speed, cost and numbers. Their enablers were: commercial electronics (processors, cameras, radios) that became small, capable and cheap; standardised interfaces and deployers; inexpensive rideshare launch; and a culture of rapid iteration ("fail fast"). Figure 11.9 shows how launches have shifted over the past 15 years — first towards very small satellites, then towards mass-produced constellation satellites in the 100–2 500 kg range.



Data: GCAT. Starlink V1/V1.5 (≈ 260–310 kg) fill the 100–500 kg class to 2022; V2 Mini (≈ 800 kg) and Amazon Leo (≈ 600 kg) fill the 500–2 500 kg class from 2023.

FIGURE 11.9 Payloads launched per year by launch mass, 2010–2025 (GCAT). The explosion after 2019 is driven by mass-produced constellation satellites; CubeSats (< 10 kg) continue at several hundred per year.

11.7.2 The CubeSat Standard

The **CubeSat** was proposed in 1999 by Jordi Puig-Suari (California Polytechnic State University) and Bob Twiggs (Stanford) as a standard for student satellites. It is a 10 cm cube ("1U") of up to about 2 kg. It is launched from a standard spring-loaded box deployer (the P-POD and its successors) that protects the launch vehicle and primary payload. Larger formats stack units: 3U, 6U, 12U and 16U are now the most common (Figure 11.10). The standard's genius is that it decouples the satellite from the launcher: any CubeSat fits any compatible deployer on any rocket or on the ISS.

CubeSat Design Specification (Cal Poly, rev. 14): 1U = 10 cm cube; rails slide in a spring-loaded deployer

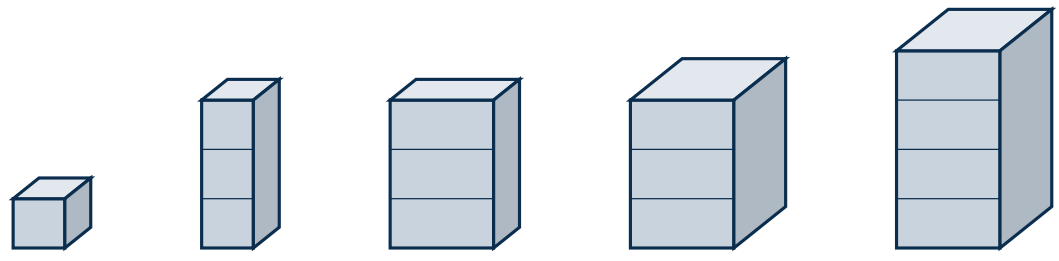


FIGURE 11.10 Common CubeSat form factors and approximate maximum masses.

A thriving commercial supply chain now sells CubeSat buses and components "off the shelf". These include structures, solar panels, batteries and power units, reaction wheels and star trackers. There are also flight computers running Linux or FreeRTOS, S- and X-band radios, and even electric thrusters. A 6U Earth-observation CubeSat can be built and launched for a few million dollars in under two years.

11.7.3 What Small Satellites Do

- **Earth observation** — Planet's fleet of about 200 "Dove" and "SuperDove" 3U CubeSats images the entire land surface daily at 3–5 m resolution; larger SkySats and Pelican satellites offer 50 cm imaging.
- **Radio-frequency data** — Spire's Lemur CubeSats collect GNSS radio-occultation weather data, ship AIS and aircraft ADS-B signals; HawkEye 360 geolocates RF emitters (Chapter 19).
- **Radar** — ICEYE, Capella and Umbra fly 50–150 kg SAR microsatellites (Chapter 15).
- **Science** — NASA's TROPICS CubeSats measure tropical-cyclone microwave signatures every hour; GRACE-like and space-weather missions use CubeSat swarms.
- **Deep space** — NASA's twin **MarCO** CubeSats relayed InSight's landing data from Mars in 2018 — the first CubeSats beyond Earth orbit. Ten CubeSats rode on Artemis I to the Moon (2022), and CAPSTONE pioneered the Gateway's NRHO.
- **Technology demonstration and education** — thousands of students worldwide have built CubeSats; the ISS deploys many of them.

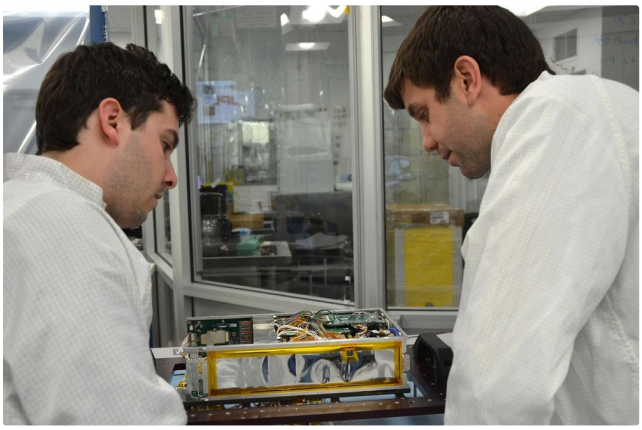


FIGURE 11.11 JPL engineers with one of the twin MarCO 6U CubeSats, which flew alongside InSight to Mars in 2018 and relayed its landing data in real time. Image: NASA/JPL (Elizabeth Landau).



FIGURE 11.12 NASA astronaut Frank Rubio installs a NanoRacks CubeSat Deployer in the ISS Kibo airlock (2023). Deployers are passed outside through the airlock and the station's robotic arm releases the CubeSats. Image: NASA/JSC.

11.7.4 Reliability of Small Satellites

Early CubeSats had high failure rates. Studies of missions launched up to the mid-2010s found that roughly 20–40 % failed early in their missions, often due to power, communications or deployment failures. University "first missions" fared worst. Reliability has improved greatly with commercial heritage hardware, more thorough testing ("flatsat" functional testing, thermal-vacuum and vibration) and better operations. Commercial constellations accept some individual failures because the system degrades gracefully; science and defence small satellites adopt selected high-reliability practices ("Class D" risk posture, Chapter 4).

11.8 Launch and Early Operations

The first days after launch — the **launch and early orbit phase (LEOP)** — are the riskiest of a satellite's life. A typical LEO satellite's LEOP sequence is:

1. **Separation** from the launch vehicle; the spacecraft wakes on a separation switch, powers essential units and starts timers.
2. **Detumbling** — rates from separation (up to a few degrees per second) are damped using magnetorquers ("B-dot" control) or thrusters.
3. **Sun acquisition** — solar arrays are deployed (if folded) and pointed at the Sun to make the spacecraft power-positive.
4. **First contact** — ground stations around the world, often including commercial network stations, search for the beacon. Orbit determination from the launcher's injection data and early ranging refines the orbit; for rideshare launches, operators must first identify which of dozens of newly catalogued objects is theirs.
5. **Deployments** — antennas, booms, radiator covers and instrument doors are released.
6. **Commissioning** — each subsystem and the payload are switched on and verified; propulsion is tested with small burns; the satellite is manoeuvred to its operational orbit.
7. **Calibration and validation** of the payload, typically lasting weeks (communications) to months (science instruments), before the satellite enters routine operations.

11.9 Orbit Maintenance

Satellites in LEO must counter drag to stay in their slots; constellations must also maintain their relative phasing. The velocity lost to drag each year is the integral of the drag deceleration over the year:

$$\Delta v_{year} = \int_{1\,\mathrm{yr}} \frac{\rho v^2}{2B}\, dt \;\approx\; \frac{\rho v^2}{2B} \times 3.16 \times 10^{7}\ \mathrm{s}$$

At 550 km with moderate solar activity ($\rho \approx 5 \times 10^{-13}$ kg/m^3, v = 7.6 km/s) and B = 50 kg/m^2, the drag deceleration is $5 \times 10^{-13} \times (7\,600)^2/100 \approx 2.9 \times 10^{-7}$ m/s^2. This is about **9 m/s per year**. At 400 km and solar maximum it can exceed 100 m/s per year.

11.10 A Worked Small-Satellite Design

To see how the material of Part I comes together, consider the preliminary design of a 12U CubeSat for agricultural monitoring.

Requirements. Multispectral images (blue, green, red, red-edge, near-infrared) at 5 m ground sampling distance (GSD) over Europe, revisit ≤ 5 days with a constellation, 3-year life, launch on a rideshare to SSO.

Orbit. 525 km sun-synchronous (97.5°), 10:30 local time. Lifetime analysis (Chapter 2) gives 6–15 years for a 12U CubeSat with deployed panels, so a small thruster for collision avoidance and end-of-life lowering is advisable to meet the 5-year disposal rule with margin.

Payload optics. GSD = altitude × pixel pitch / focal length. With 5.5 µm pixels: $f = 525\,000 \times 5.5 \times 10^{-6}/5 = 0.58$ m. A 9 cm-aperture compact catadioptric telescope with a 580 mm focal length fits in about 4U. At f/6.4 the diffraction limit at 800 nm ($2.44\lambda N \approx 12.5$ µm spot diameter) is acceptable for 5 m GSD, with some MTF loss. A 4 096-pixel line array gives a 20 km swath.

Data budget. Imaging Europe's croplands ≈ 2 million km² per 5 days per satellite with 5 bands × 12 bits at 5 m: $2 \times 10^6\,\text{km}^2/5\,\text{days} \times (10^6/25)\,\text{pixels/km}^2 \times 60\,\text{bits} \approx 1 \times 10^{12}$ bits/day ≈ 1 Tbit/day before compression. With 3:1 compression and onboard cloud screening (Chapter 45) removing about half, the downlink requirement is ≈ 170 Gbit/day. An X-band transmitter at 200 Mbit/s needs about 14 minutes of contact per day — two or three passes over a high-latitude station.

Power budget.

Mode	Power (W)	Duty / orbit	Energy per orbit (Wh)
Platform (OBC, ADCS, receivers, heaters)	12	100 %	19.1
Imaging (payload + ADCS fine pointing)	25	10 %	4.0
X-band downlink	30	7 %	3.3
Total	—	—	≈ 26.4 per 95.3-min orbit

Deployable panels with triple-junction cells (≈ 80 W peak, ≈ 45 W orbit-average after pointing losses) give a positive balance with 30 % margin. A 90 Wh Li-ion battery supports eclipse at < 20 % depth of discharge.

Mass budget.

Item	Mass (kg)
Structure and deployers	3.2
Payload telescope and focal-plane electronics	5.5
Power (panels, battery, EPS)	3.0
ADCS (3 wheels, star tracker, magnetorquers, sensors)	1.6
OBC and payload processor (with AI accelerator)	0.6
S-band TT&C and X-band transmitter with patch array	0.9
Propulsion (electrospray or cold gas)	1.2
Harness, thermal hardware	0.8
Subtotal	**16.8**
Margin (25 %)	4.2
Total (12U limit ≈ 24 kg)	**21.0**

Constellation. With a 20 km swath, one satellite cannot meet a 5-day revisit over Europe by nadir imaging alone. Pointing ±20° off-nadir gives an accessible swath of about 400 km. So a constellation of four to six satellites in one or two planes meets the requirement. This example illustrates how requirements cascade through optics, data, power, mass and constellation size — the essence of systems engineering (Chapter 4).

11.11 Getting Small Satellites to Orbit

11.11.1 Rideshare

Most small satellites fly as **secondary payloads** sharing a launch with others. SpaceX's **Transporter** (sun-synchronous) and **Bandwagon** (mid-inclination) missions carry 50–120 satellites per flight from many customers; ISRO's PSLV, Rocket Lab's Electron, Arianespace's Vega-C (SSMS dispenser) and China's Long March and commercial rockets offer similar services. Rideshare is cheap (≈ US$6 000–7 000 per kg on Transporter in 2025) but the customer does not choose the orbit or the date. **Dedicated small launchers** (Electron, Firefly Alpha, Kinetica-1, Ceres-1, Zhuque-2, Vega-C, PSLV, SSLV) provide orbit and schedule control at higher cost per kilogram (Chapter 36).

11.11.2 Orbital Transfer Vehicles

Orbital transfer vehicles (OTVs) or "space tugs" bridge the gap: they ride a rideshare launch, then carry their satellite passengers to individual orbits or planes. Examples include D-Orbit's ION Satellite Carrier (more than 15 missions since 2020), Exotrail's spacevan, Momentus' Vigoride, Impulse Space's Mira, Rocket Lab's Photon and Firefly's Elytra. Some OTVs also host experiments and are evolving into servicing vehicles.

11.11.3 Deployment from the ISS

The ISS has deployed hundreds of CubeSats through the Japanese **J-SSOD** and commercial NanoRacks deployers in the Kibo module airlock. ISS-deployed satellites start at about 410 km and decay within one to three years — an advantage for debris mitigation.

11.12 Formation Flying and Swarms

Some missions require two or more satellites to fly in a controlled geometry:

- **GRACE and GRACE-FO** (US–Germany) — two satellites 220 km apart in the same orbit measure their separation to micrometres (GRACE-FO adds a laser ranging interferometer at nanometre precision). These measurements map changes in Earth's gravity from melting ice, groundwater and ocean mass.
- **TanDEM-X** (Germany) — two radar satellites flying in a close "helix" formation a few hundred metres apart acted as a single-pass interferometer to produce a global digital elevation model (Chapter 15).
- **PROBA-3** (ESA, launched December 2024) — two satellites flying about 150 m apart in a highly elliptical orbit. One acts as an occulting disc for the other's coronagraph to create artificial solar eclipses, with millimetre-level relative positioning. It produced its first artificial-eclipse images of the solar corona in 2025.
- **MMS** (NASA) — four spacecraft in a tetrahedral formation measuring magnetic reconnection (Chapter 23).
- **Starling** (NASA, 2023) — four CubeSats demonstrating autonomous swarm manoeuvre planning and relative navigation.

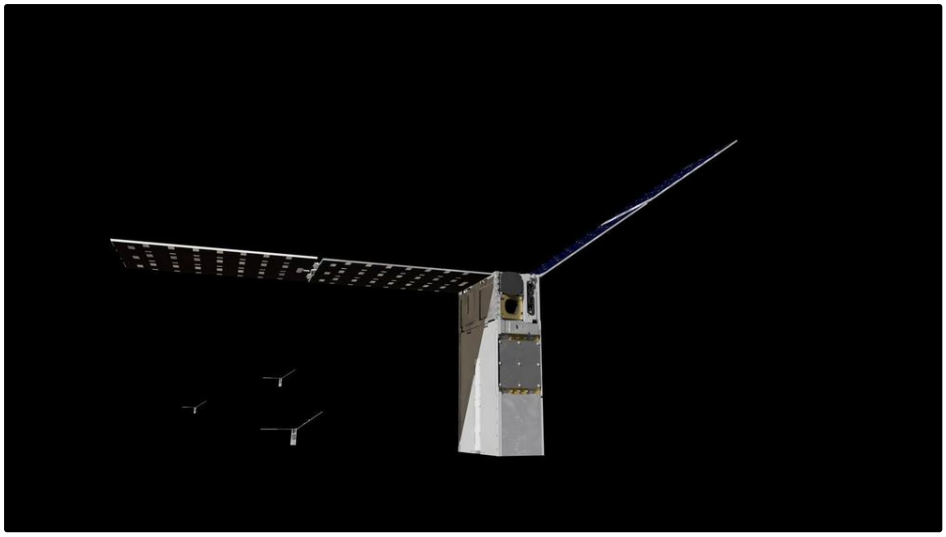


FIGURE 11.13 Illustration of NASA's Starling mission: four CubeSats flying as an autonomous swarm in low Earth orbit (2023–2024). Image: NASA/ARC (NASA).

Formation flying requires precise relative navigation (carrier-phase differential GNSS, inter-satellite radio or optical ranging, vision sensors) and control using small thrusters or differential drag. Future concepts include distributed-aperture telescopes and interferometers, synthetic-aperture radar clusters, and gravitational-wave observatories (LISA, Chapter 39).

11.13 Operating Large Constellations

Operating thousands of satellites is impossible with traditional mission control, where a team watches each spacecraft. Constellation operators rely on:

- **Automation** — automated planning and scheduling of ground contacts, autonomous orbit maintenance and collision avoidance, automated anomaly detection and recovery; humans manage by exception.
- **Fleet management** — configuration management of thousands of identical vehicles, software updates over the air, health trending with machine learning (Chapter 49).
- **Ground networks** — hundreds of gateways (Starlink operates more than 150 gateway sites) and cloud-based processing.
- **Replenishment** — continuous launches to replace satellites that reach end of life (typically after five years in LEO) and to upgrade technology.
- **Space traffic coordination** — sharing ephemerides and manoeuvre plans with other operators and with space-surveillance organisations (Chapter 21).

11.14 Sustainability, Regulation and the Night Sky

The rapid growth of constellations raises four sustainability issues:

1. **Orbital debris and collision risk.** Large numbers of satellites increase conjunctions. Mitigation measures include low operating altitudes, reliable propulsion and prompt disposal. The US FCC requires LEO satellites ending their missions to deorbit within **5 years**, a

rule adopted in 2022 that replaced the older 25-year guideline. Other measures are high post-mission disposal reliability (> 90–99 %) and design for demise (Chapter 21).

2. **Spectrum.** The ITU requires non-geostationary systems to deploy **10 % of their filed satellites within 2 years**, **50 % within 5 years** and **100 % within 7 years** after the end of the regulatory period. These rules were adopted at WRC-19. Systems must meet them to keep their priority — driving the "race" to launch. Interference with GEO systems is limited by equivalent power-flux-density (EPFD) rules.
3. **Astronomy.** Sunlit satellites appear as moving streaks in telescope images, particularly in twilight and in wide-field surveys such as the Vera C. Rubin Observatory's Legacy Survey of Space and Time. Operators have reduced brightness with dark coatings, visors and attitude changes, and the International Astronomical Union's Centre for the Protection of the Dark and Quiet Sky coordinates mitigation. Unintended radio emissions from satellite electronics also affect radio astronomy.
4. **Atmosphere.** Re-entering satellites (several per day from large constellations) deposit aluminium oxide and other materials in the upper atmosphere, with possible effects on ozone chemistry that are under study.

KEY TAKEAWAYS

- Orbit choice sets coverage, revisit, resolution, latency, environment and cost; coverage geometry links altitude, minimum elevation and footprint.
- Walker delta and polar "street-of-coverage" constellations are the classic designs; broadband constellations need thousands of satellites for capacity, not just coverage.
- In October 2026 about 35 000 objects were catalogued in Earth orbit (≈ 17 200 operational payloads). More than half of all objects orbit between 450 and 600 km, dominated by Starlink (> 10 600 satellites).
- Mega-constellations (Starlink, OneWeb, Amazon Leo, Qianfan, Guowang, IRIS^2, SDA layers) are reshaping the industry and the orbital environment.
- GEO is a limited resource managed through ITU slots and station-keeping boxes, now evolving towards software-defined and smaller satellites.
- CubeSats and small satellites, rideshare launches and orbital transfer vehicles made space accessible to universities, start-ups and new nations; sustainability rules (5-year disposal, ITU milestones, brightness mitigation) are evolving with them.

REVIEW QUESTIONS

1. List six mission characteristics that depend on the choice of orbit.
2. A satellite at 550 km altitude serves users down to 25° elevation. Calculate the Earth central angle and the ground radius of its coverage circle.
3. Explain Walker delta notation i: $T/P/F$, using a GNSS constellation as an example.
4. Why do broadband constellations need thousands of satellites when a few dozen could cover the whole Earth?
5. How are satellites launched into one plane redistributed into several planes without large plane-change manoeuvres?
6. When do geostationary satellites experience eclipses, and how long can an eclipse last?
7. What features of the CubeSat standard made small satellites affordable? What are the main reliability concerns?

8. Describe three rules or practices that aim to keep low Earth orbit sustainable and limit the effect of constellations on astronomy.

CHAPTER

12 Satellite Buses and Payload Instruments



Image: NASA/KSC (Michael Starobin NOAA NASA)

Every satellite is a payload carried by a bus. This chapter surveys the commercial and government satellite platforms on which payloads fly - from standard geostationary buses to mass-produced constellation platforms and CubeSat kits - and the interfaces between bus and payload. It then introduces the families of payload instruments that later chapters apply: optical and infrared imagers and spectrometers, microwave radiometers, radars, lidars, particle and field instruments and service payloads, with the physics of detectors, telescopes, scanning methods, calibration and the payload data chain.

IN THIS CHAPTER

12.1 The Bus as a Product
12.2 A Taxonomy of Payload Instruments
12.3 Optical and Infrared Instruments
12.4 Microwave Instruments
12.5 Lidar Instruments
12.6 In-Situ Instruments: Fields and Particles
12.7 Service Payloads
12.8 Performance, Calibration and Data
12.9 Designing an Optical Payload: Radiometry and SNR
12.10 Instrument Case Studies
12.11 Large Communications Platforms: Power as the Measure

12.1 The Bus as a Product

12.1.1 From Custom Spacecraft to Product Lines

Early satellites were built one at a time around their payloads. Today most satellites use a **standard bus** (platform) from a manufacturer's product line, adapted to the payload — much as a truck chassis carries different bodies. Product lines lower cost and risk through heritage (flight-proven designs), reuse of qualification data, and streamlined production. Table 12.1 lists representative product lines.

TABLE 12.1 Representative satellite bus product lines (2026).

Class	Bus	Manufacturer	Typical mass / power	Typical missions
Large GEO	Eurostar Neo; OneSat (software-defined)	Airbus (Europe)	3–6.5 t / up to 25 kW	Broadband and broadcast comsats

Class	Bus	Manufacturer	Typical mass / power	Typical missions
Large GEO	Spacebus Neo; Inspire	Thales Alenia Space (Europe)	3–6 t / up to 20 kW	Comsats, Galileo second generation (with Airbus)
Large GEO	702 family (702SP, 702X)	Boeing (USA)	2–6 t / 3–25 kW	Comsats, WGS, O3b mPOWER (MEO)
Large GEO	LM 2100 / LM 400	Lockheed Martin (USA)	2–6 t	GOES-R, GPS III (A2100-derived), SBIRS, comsats
Large GEO	1300 series	Maxar Space Systems (USA)	3–7 t / up to 25 kW	Comsats, Psyche (derived bus)
Large GEO	DFH-4 / DFH-4E / DFH-5	CAST (China)	5–9 t / up to 28 kW	Chinese comsats, Tianlian relays
Large GEO	I-3K / I-4K / I-6K	ISRO (India)	2–6 t	GSAT, INSAT
Small GEO	MicroGEO	Astranis (USA)	≈ 400 kg	Regional broadband
LEO / MEO medium	Eurostar / AstroBus / Arrow	Airbus	0.3–2 t	Sentinel, EO, OneWeb (Arrow)
LEO medium	SSTL-150 / -300 / -S1	Surrey Satellite Technology (UK)	100–500 kg	EO, Galileo FOC payloads, technology
LEO medium	LM 400 / LM 50	Lockheed Martin	50–2 000 kg	SDA, military
LEO productised	Aries / Nova / Comet	Apex (USA)	100–500 kg	Rapid-delivery buses for many customers
LEO productised	X-Bus, MicroSat	York Space / Blue Canyon (RTX)	50–500 kg	SDA Transport Layer, NASA science (TROPICS)
Constellation-specific	Starlink V2 Mini / V3	SpaceX	0.8–2 t	Vertical integration, in-house only
CubeSat kits	3U–16U buses	Blue Canyon, EnduroSat, GomSpace, NanoAvionics, AAC Clyde, Open Cosmos	4–30 kg	EO, IoT, science, education

Three major trends are reshaping bus design:

1. **Productisation and speed.** Companies such as Apex and York aim to deliver standard buses in months rather than years, with large factories building dozens per year. Governments (e.g., the US Space Development Agency) buy satellites in “tranches” of tens to hundreds.
2. **Software-defined platforms and payloads.** Digital processors and flexible antennas let a satellite’s coverage, frequency plan and power allocation be changed in orbit — and let the bus’s own behaviour be updated like a phone’s operating system.
3. **Vertical integration.** The largest constellation operators build their own buses, payloads, ground terminals and launches, optimising the whole system rather than buying components.

12.1.2 What the Bus Provides

From the payload’s point of view, the bus is a service provider. It must deliver:

- **A place to live**: mechanical mounting with the required alignment and stiffness, protection during launch.
- **Power**: average and peak power at the right voltage, with protected switching.
- **Thermal environment**: heat removal for electronics, stable temperatures for optics, cooling for detectors.
- **Pointing**: accuracy, knowledge, stability (jitter), agility and keep-out zones (e.g., never point a sensitive telescope near the Sun).
- **Data handling**: command distribution, data collection, storage and downlink capacity.
- **Time and position**: precise time (often a GNSS-derived pulse per second) and orbit/attitude data for geolocation.

- **A clean environment**: low contamination, low electromagnetic interference and low microvibration.

These services are defined in a **payload–bus interface control document** (Figure 12.1). Mismatches at interfaces — a payload drawing more peak power than expected, or a bus reaction wheel exciting a resonance of the telescope — are among the commonest causes of problems in integration.

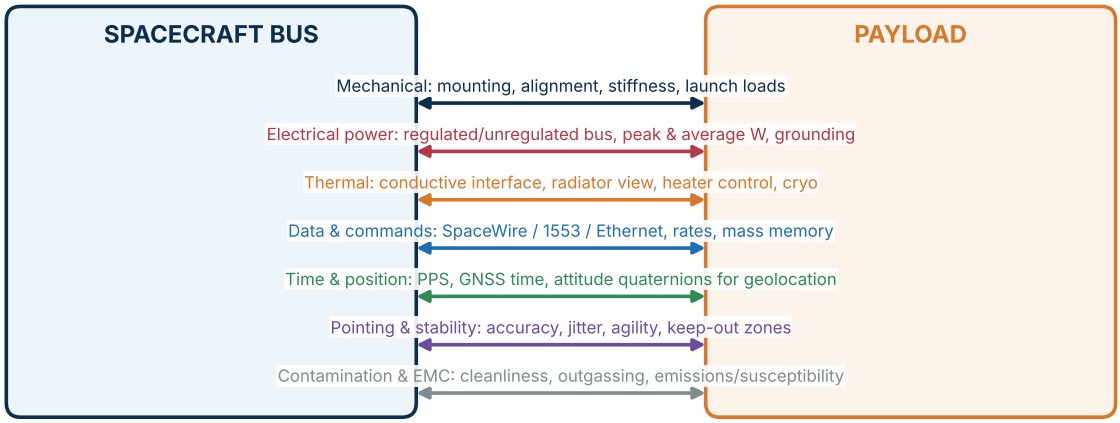


FIGURE 12.1 The payload–bus interfaces that must be specified and verified for every mission.

12.1.3 Hosted Payloads

A **hosted payload** is an instrument or transponder that flies on someone else's satellite, using spare mass, power and data capacity — typically on a commercial GEO communications satellite. Hosting can cut costs by a factor of several and speed access to orbit. Examples include NASA's GOLD ultraviolet imager (on SES-14, 2018) and TEMPO air-quality spectrometer (on Intelsat 40e, 2023). Others are the Aireon ADS-B receivers on all 66 Iridium NEXT satellites (giving global air-traffic surveillance), and several US and European government payloads on commercial satellites. The challenges are aligning schedules, defining interfaces and ensuring that the host's operations do not compromise the hosted mission.

12.2 A Taxonomy of Payload Instruments

Payloads can be grouped in four families (Figure 12.2):

- **Passive remote sensing** — instruments that measure radiation emitted or reflected by the target: cameras, imagers, spectrometers, radiometers, polarimeters, and receivers of GNSS signals (radio occultation and reflectometry).
- **Active remote sensing** — instruments that transmit their own signal and measure its echo: radars (synthetic aperture radar, altimeters, scatterometers, precipitation and cloud radars) and lidars.
- **In-situ instruments** — instruments that measure their immediate environment: magnetometers, plasma and particle detectors, electric-field and wave instruments, dust detectors, mass spectrometers and accelerometers.
- **Service payloads** — communications transponders, navigation payloads (clocks and signal generators), data relay terminals, search-and-rescue repeaters.

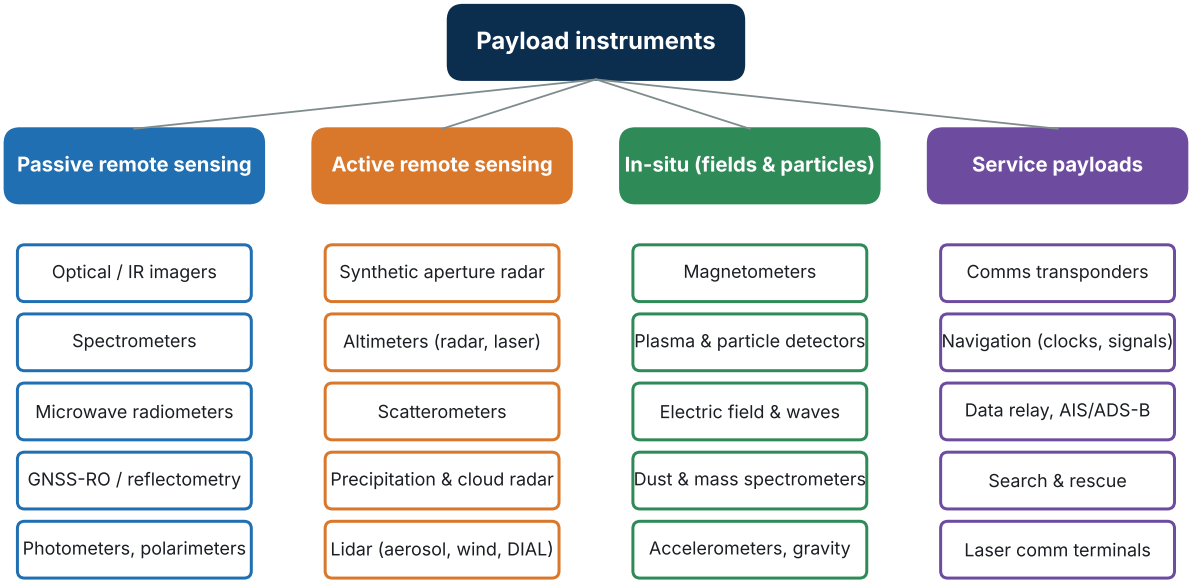


FIGURE 12.2 A taxonomy of satellite payloads.

Each part of the electromagnetic spectrum carries different information and needs different technology (Figure 12.3). Short wavelengths (X-rays, gamma rays) reveal high-energy processes, and ultraviolet senses ozone, the upper atmosphere and hot plasmas. Visible and near-infrared light shows surface reflectance (vegetation, minerals, water), and thermal infrared shows temperature and composition. Microwaves penetrate clouds and sense moisture, temperature profiles and surface roughness; radio wavelengths are used for radar, navigation and communications.

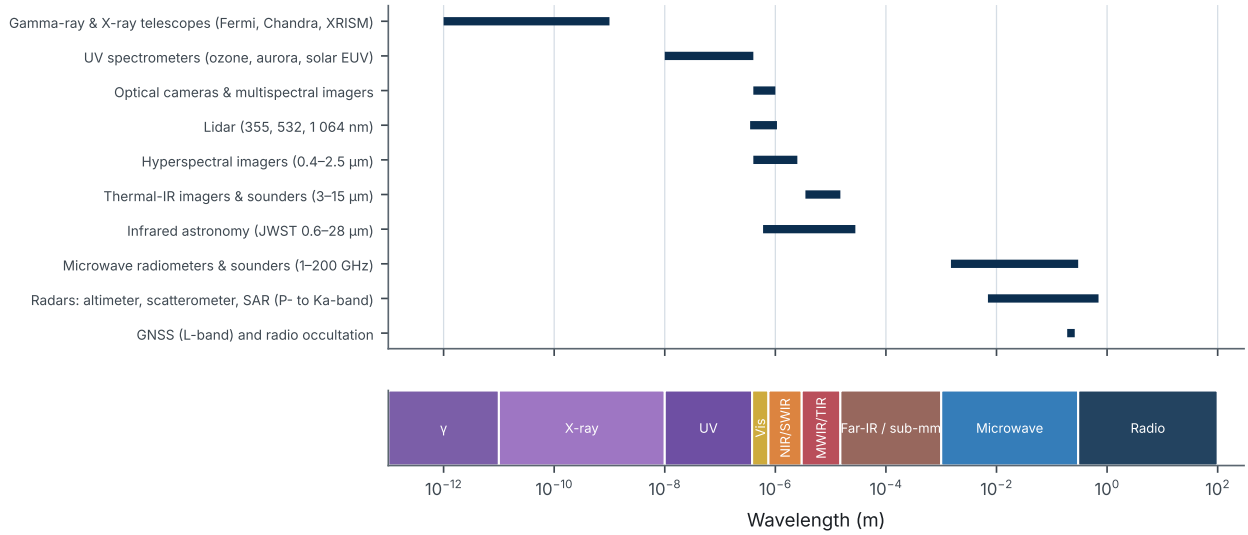
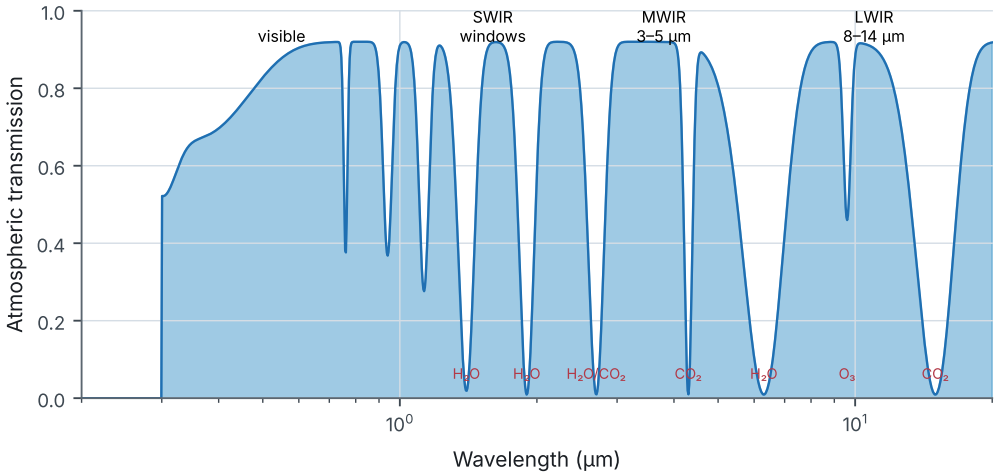


FIGURE 12.3 The electromagnetic spectrum and the wavelength ranges used by different classes of space instruments.

Not all wavelengths reach the ground from space: the atmosphere absorbs strongly in certain bands (Figure 12.4). Imagers that want to see the surface use **atmospheric windows**, while **sounders** deliberately use absorption bands. At different distances from the centre of a CO_2 or O_2 band, the radiation comes from different heights. This lets sounders retrieve vertical profiles of temperature and humidity (Chapter 16).



Schematic vertical transmission of a mid-latitude atmosphere. Imagers use the windows; sounders deliberately observe within absorption bands to probe different heights.

FIGURE 12.4 Schematic atmospheric transmission from 0.2 to 20 µm. Windows are used for surface imaging; absorption bands of water vapour, carbon dioxide and ozone are used for atmospheric sounding.

12.3 Optical and Infrared Instruments

12.3.1 Telescopes

An optical instrument collects light with a **telescope** and focuses it onto a detector. The main designs (Figure 12.5) are:

- **Refractors** — lenses; simple but heavy and limited in aperture; used in star trackers, small cameras and wide-field instruments.
- **Cassegrain-type reflectors** — a concave primary and convex secondary mirror; the **Ritchey–Chrétien** variant (Hubble, many Earth-imaging telescopes) corrects coma for a wider field.
- **Three-mirror anastigmats (TMA)** — three curved mirrors, often off-axis, give wide, well-corrected fields with no central obstruction; used in pushbroom imagers (Landsat OLI, Sentinel-2 MSI, WorldView) and JWST.
- **Segmented and deployable primaries** — large telescopes that fold for launch (JWST's 6.5 m, 18-segment mirror; Chapter 38).
- **Catadioptric** designs combining lenses and mirrors, compact enough for small satellites.

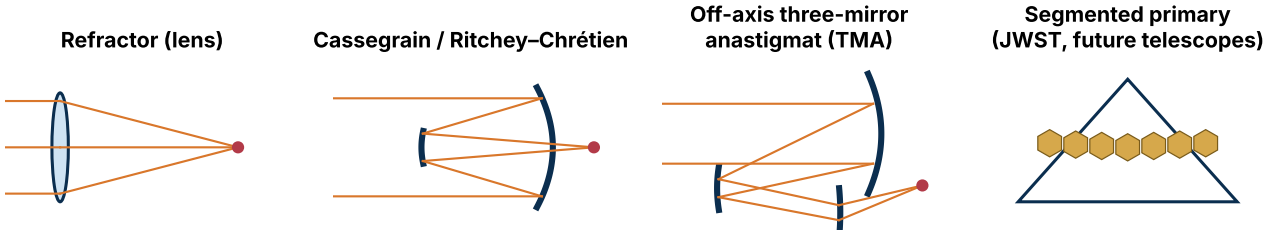


FIGURE 12.5 Telescope configurations used in space instruments.

The angular resolution of a telescope is limited by diffraction:

$$\theta \approx 1.22 \frac{\lambda}{D}$$

For a 1.1 m telescope (WorldView-3) at 0.5 µm, $\theta = 0.55$ µrad, so from 617 km it can resolve about 34 cm on the ground. Real resolution also depends on the detector pixel size, optical aberrations, satellite motion and jitter, and the atmosphere — summarised in the **modulation transfer function (MTF)** (Chapter 13).

12.3.2 Detectors

Detectors convert photons into electrical signals. Visible and near-infrared imagers use **silicon** detectors — CCDs (charge-coupled devices) and, increasingly, CMOS image sensors, which integrate readout electronics on the chip, consume less power and are more tolerant of radiation. **Time-delay integration (TDI)** sensors shift charge along columns in step with the image motion, adding up the signal from tens of rows to increase sensitivity for fast-moving LEO imagers.

Infrared detectors use semiconductors with smaller band gaps: **InGaAs** (short-wave IR to 1.7 µm, usable at near room temperature) and **InSb** (to 5.5 µm, cooled to about 80 K). **HgCdTe** ("MCT") is tunable from 1 to more than 14 µm by changing its composition; it is the workhorse of space IR instruments, including JWST's NIRCam and NIRSpec. Newer types are **type-II superlattice** (T2SL) and **barrier** detectors. **Microbolometers** detect the heating of tiny suspended membranes and need no cooling, making thermal cameras possible on small satellites, although with lower sensitivity (Figure 12.6).

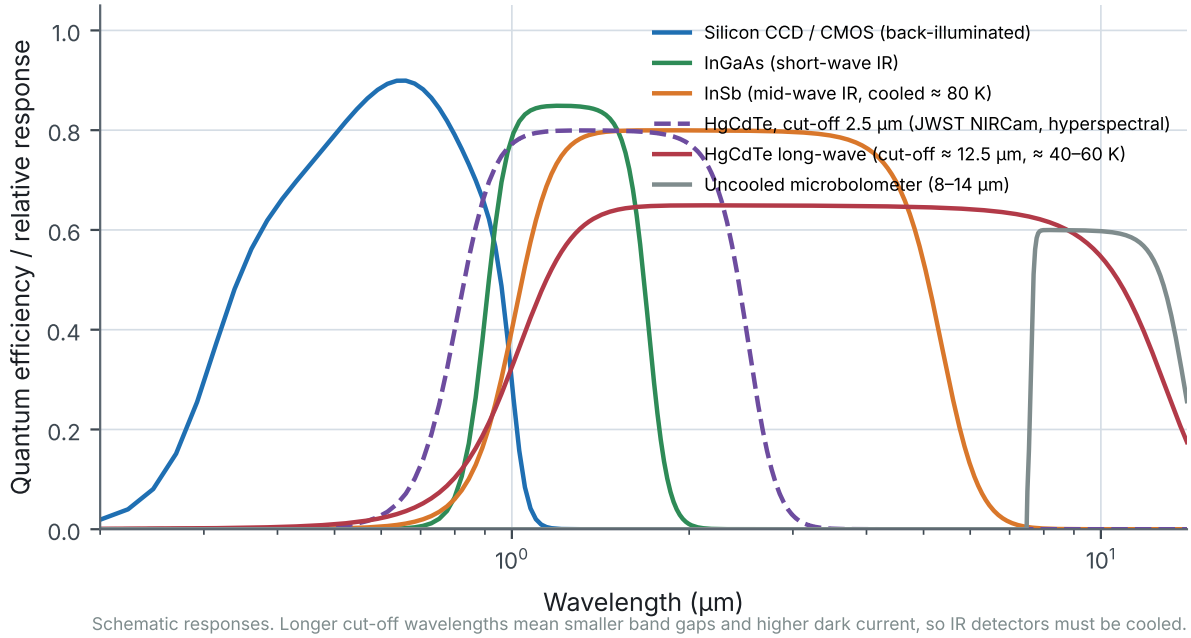


FIGURE 12.6 Schematic quantum efficiency or relative response of common detector materials. Each material covers a spectral range set by its band gap; infrared detectors must be cooled to suppress dark current.

The dark current of a detector grows exponentially with temperature and inversely with band gap. So long-wave infrared detectors must be cooled — to 40–80 K for HgCdTe and InSb, and to below 10 K for some astronomical detectors. Cooling uses **passive radiators** (for missions with cold views, such as polar orbiters and L2 telescopes) or **cryocoolers** (Chapter 5).

12.3.3 Imaging Architectures

Imagers build a 2-D picture of the ground in three ways (Figure 12.7):

- **Whiskbroom** scanners use a rotating or oscillating mirror to sweep a few detectors across the swath, building the image line by line as the satellite moves (Landsat 1–7 MSS/TM/ETM+, AVHRR, MODIS). They need few detectors but have short dwell times and moving parts.
- **Pushbroom** imagers use long linear arrays (thousands to tens of thousands of pixels) perpendicular to the ground track; the satellite's motion sweeps the array along. They have long dwell time, no scanning mirror and excellent geometry — the standard for modern EO (Landsat 8/9 OLI, Sentinel-2 MSI, WorldView, Pléiades).
- **Framing** (staring) cameras use 2-D arrays to capture whole scenes at once, like a normal camera; used on SkySat, Planet Doves, many GEO imagers in step-stare mode, and astronomical telescopes. **Video** from orbit is possible with framing cameras on agile satellites.

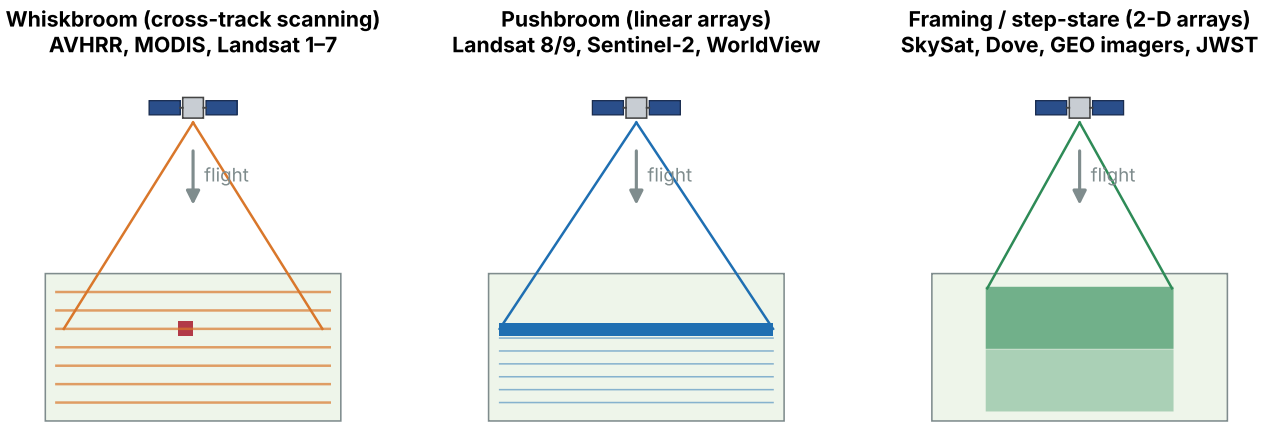


FIGURE 12.7 The three basic imaging architectures.

12.3.4 Spectrometers

A **spectrometer** splits light into its component wavelengths to measure composition. Main types:

- **Dispersive** spectrometers use a **prism** or **diffraction grating**. An imaging spectrometer forms a 2-D detector image with one axis spatial and one axis spectral (hyperspectral pushbroom imagers such as EMIT, PRISMA and EnMAP; Chapter 14).
- **Fourier-transform spectrometers (FTS)** measure an interferogram with a moving mirror (Michelson interferometer) and transform it into a spectrum with high spectral resolution; used by infrared sounders (IASI, CrIS) and greenhouse-gas instruments (GOSAT).
- **Filter-based** spectral imagers use fixed or tunable filters (multispectral imagers; Fabry–Pérot and acousto-optic tunable filters).
- **Spatial heterodyne** and **grating-based high-resolution** spectrometers for specific gases (OCO-2/3 measure CO_2 with three grating spectrometers at 0.76, 1.61 and 2.06 μm).

12.4 Microwave Instruments

12.4.1 Passive Microwave Radiometers

Every object emits microwave radiation in proportion to its temperature and **emissivity**. In the microwave region the Rayleigh–Jeans approximation holds, so the measured radiance is expressed as a **brightness temperature** $T_B = \varepsilon T_{physical}$. Radiometers measure T_B at several frequencies and polarisations to retrieve:

- **Sea-surface temperature and wind speed** (6–37 GHz; AMSR2, GMI);
- **Soil moisture and ocean salinity** at L-band (1.4 GHz; SMOS, SMAP), where emissivity depends strongly on water content;
- **Sea ice, snow and rain** (10–89 GHz);
- **Atmospheric temperature profiles** using the 50–60 GHz oxygen band and **humidity profiles** using the 183 GHz water-vapour line (ATMS, AMSU, MWS on MetOp-SG); see Chapter 16.

Because microwave wavelengths are long, antennas must be large for fine resolution. At 1.4 GHz a 6 m reflector (SMAP) gives about 40 km resolution. SMOS used a 2-D **interferometric** array of 69 small antennas to synthesise a large aperture. Radiometers are calibrated on every scan by viewing a warm blackbody target and cold space.

12.4.2 Radars

Active microwave instruments transmit pulses and analyse the echoes:

- **Synthetic aperture radar (SAR)** — high-resolution imaging day and night through cloud, using the satellite's motion to synthesise a long antenna (Chapter 15).
- **Radar altimeters** — measure the height of the ocean, ice or land surface to centimetre precision from the round-trip time of nadir pulses (Jason, Sentinel-3, Sentinel-6 Michael Freilich). **SWOT** (2022) uses Ka-band radar interferometry to map water heights in 2-D swaths.
- **Scatterometers** — measure backscatter from the ocean surface from several directions to infer wind speed and direction (ASCAT on MetOp, HY-2 series).
- **Precipitation radars** (Ku/Ka-band; GPM's Dual-frequency Precipitation Radar) and **cloud radars** (94 GHz; CloudSat, EarthCARE's Doppler cloud profiling radar).
- **Sounding radars** — low-frequency radars that penetrate ice and dry ground (MARSIS and SHARAD at Mars, RIME on JUICE, REASON on Europa Clipper).

12.4.3 GNSS Remote Sensing

Navigation signals are a free, global source of microwave illumination. In **radio occultation (RO)**, a LEO receiver tracks a GNSS signal as it sets behind the Earth's atmosphere. The bending of the signal gives profiles of refractivity, hence temperature and humidity, with high vertical resolution and no calibration drift. These profiles are among the most valuable inputs to weather forecasting (COSMIC-2, MetOp GRAS, Spire, PlanetiQ). In **GNSS reflectometry (GNSS-R)**, receivers measure signals reflected from the ocean or land to infer wind speed, soil moisture and flooding (CYGNSS, 2016).

12.5 Lidar Instruments

A **lidar** (light detection and ranging) fires laser pulses and measures the return. Space lidars include:

- **Altimetry lidars** — ICESat (2003–2009) and **ICESat-2** (2018–). ICESat-2's ATLAS instrument fires 10 000 pulses per second split into six beams, and counts individual returning photons to measure ice-sheet and sea-ice elevation to centimetres. **GEDI** on the ISS measures forest canopy height.
- **Backscatter lidars** — CALIPSO (2006–2023) and EarthCARE's ATLID (2024) profile aerosols and clouds, separating particle types by depolarisation.
- **Doppler wind lidar** — ESA's **Aeolus** (2018–2023) measured wind profiles globally for the first time from the Doppler shift of molecular and aerosol backscatter at 355 nm, improving weather forecasts.
- **Differential absorption lidar (DIAL)** — measures gases such as methane (France–Germany's planned MERLIN) by comparing absorption at two nearby wavelengths.
- **Planetary lidars** — the Mars Orbiter Laser Altimeter (MOLA) and Lunar Orbiter Laser Altimeter (LOLA) mapped topography; lidars guide landers (Chapter 27).

Space lidars need powerful, efficient, long-life lasers (diode-pumped Nd:YAG at 1 064 nm and its harmonics at 532 and 355 nm; fibre lasers at 1.5–2 μm). They also need large receiving telescopes and sensitive detectors (photomultipliers, avalanche photodiodes, single-photon counters). Laser-induced damage of optics in vacuum and laser lifetime were major challenges for Aeolus.

12.6 In-Situ Instruments: Fields and Particles

Space-physics missions (Part VII) and planetary missions carry instruments that sample their surroundings directly:

- **Magnetometers** — **fluxgate** vector magnetometers (accurate to ≈ 0.01–0.1 nT) and **scalar** magnetometers (optically pumped helium, Overhauser) that measure field magnitude absolutely. They are mounted on long booms (5–10 m) to escape the spacecraft's own magnetic field, and missions follow strict **magnetic cleanliness** programmes. The ESA **Swarm** constellation maps Earth's field; Juno's magnetometers revealed Jupiter's complex dynamo.

- **Plasma analysers** — electrostatic analysers that measure the energy, direction and (with time-of-flight sections) mass of ions and electrons of eV–tens of keV.
- **Energetic particle detectors** — solid-state silicon telescopes and scintillators for MeV particles in radiation belts and solar events.
- **Electric-field and plasma-wave instruments** — long wire booms (tens of metres) and search-coil magnetometers.
- **Langmuir probes** — measure plasma density and temperature.
- **Dust detectors and mass spectrometers** — analyse cometary dust, planetary atmospheres and exospheres (e.g., Rosetta's COSIMA and ROSINA, Europa Clipper's SUDA and MASPEX).
- **Accelerometers** — measure non-gravitational accelerations for gravity missions (GOCE, GRACE-FO) and atmospheric density.



FIGURE 12.8 The Parker Solar Probe magnetometer boom being tested for deployment before launch (2018). Booms separate sensitive magnetometers from the spacecraft's own magnetic fields. Image: NASA/KSC (NASA/Leif Heimbold).

FIGURE 12.9 Europa Clipper's 8.5 m magnetometer boom, a coilable lattice mast, during deployment testing at JPL (2021). Image: NASA/JPL (NASA/JPL-Caltech).

12.7 Service Payloads

Communications payloads — transponders that receive, amplify, frequency-convert and retransmit signals, and modern regenerative payloads with digital processors and beam-forming networks — are described in Chapter 17. **Navigation payloads** — atomic clocks, signal generators and high-stability transmitters — are described in Chapter 18. Other service payloads include **search-and-rescue repeaters** (Cospas-Sarsat MEOSAR on GPS, Galileo, GLONASS and BeiDou satellites). These relay 406 MHz distress beacons and, on Galileo, send a "return link" acknowledgement. Further examples are **AIS and ADS-B receivers** for ship and aircraft tracking, **data collection systems** (Argos for animal tracking and buoys), and **laser communication terminals**.

12.8 Performance, Calibration and Data

12.8.1 Key Performance Measures

Instrument performance is captured by a small set of figures of merit:

Measure	Meaning	Typical use
Ground sampling distance (GSD) / spatial resolution	Size of a pixel on the ground; ability to separate nearby objects	Imaging (0.3 m–1 km)

Measure	Meaning	Typical use
Modulation transfer function (MTF)	Contrast preserved at a given spatial frequency (sharpness)	Optical imagers (MTF at Nyquist ≈ 0.1–0.3)
Signal-to-noise ratio (SNR)	Signal relative to noise at a reference radiance	Optical/IR imagers (100–500)
Noise-equivalent temperature difference (NEΔT)	Smallest detectable temperature change	Thermal IR (0.05–0.3 K)
Noise-equivalent differential reflectance	Smallest detectable change in reflectance	Ocean colour, hyperspectral
Spectral resolution / sampling	Width of spectral channels	Hyperspectral (5–10 nm), sounders (< 0.5 cm^{-1})
Radiometric accuracy and stability	Agreement with SI-traceable truth over time	Climate records (≤ 1–3 %; ≤ 0.1 K)
Geolocation accuracy	Error in where a pixel is placed on Earth	EO (≤ 1 pixel; < 5 m for high-res)
Revisit / temporal resolution	Time between observations	EO, weather

12.8.2 Calibration

A measurement is only useful if it is accurate and comparable over time and between instruments. Calibration (Figure 12.10) starts with **pre-launch laboratory calibration** against standards traceable to national metrology institutes (NIST, NPL, PTB). It continues with **onboard references** — sunlight scattered from diffuser panels, internal lamps, blackbodies at known temperatures, views of cold space. It also uses natural targets: the **Moon** (whose reflectance is extremely stable and is modelled by the ROLO and LIME models) and stable desert and ice sites (**vicarious calibration**, e.g., the RadCalNet network). Finally, it uses **inter-calibration** against reference instruments during simultaneous observations (the WMO **Global Space-based Inter-Calibration System, GSICS**). Future "climate benchmark" missions such as NASA's CLARREO Pathfinder and ESA's TRUTHS aim to fly SI-traceable references in orbit with uncertainties ten times smaller than today's instruments.

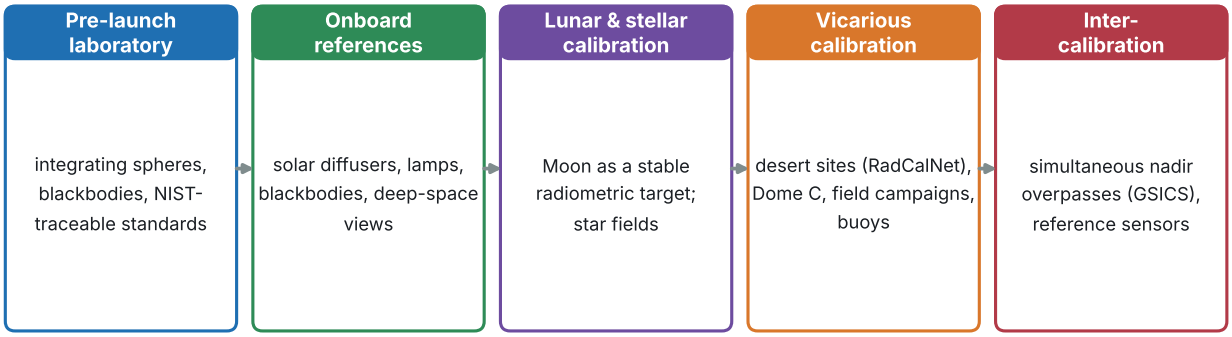


FIGURE 12.10 The calibration chain for Earth-observation instruments, from laboratory standards to in-orbit inter-calibration.

12.8.3 The Payload Data Chain

Inside the instrument, photons or radio signals become information through a chain of steps. First come the detector, analogue signal conditioning and **analogue-to-digital conversion** (typically 12–16 bits for imagers). Then come **non-uniformity correction** and binning, **compression** (lossless or near-lossless, CCSDS 121/122/123), formatting into packets with time and attitude tags, **mass memory** and downlink. Increasingly, an onboard processor or AI accelerator analyses the data before downlink — detecting clouds, ships, fires or scientific events — and sends only what matters (Chapters 45–46). On the ground, processing turns raw data (Level 0) into calibrated, geolocated products (Level 1) and geophysical variables (Level 2 and above), described in Chapter 13.

12.9 Designing an Optical Payload: Radiometry and SNR

12.9.1 How Much Light Reaches a Pixel?

The signal in an Earth-imaging detector can be estimated from the scene radiance L (W m^{-2} sr^{-1} µm^{-1}), the telescope aperture diameter D and focal length f, and the optical transmission τ. The other inputs are the spectral bandwidth $\Delta\lambda$, the detector pixel pitch p, the quantum efficiency η and the integration time t_{int}. The number of photoelectrons collected is

$$N_e = \frac{\pi}{4}\,\frac{L\,\tau\,\eta\,\Delta\lambda\,p^2\,t_{int}}{F^2}\,\frac{\lambda}{hc}, \qquad F = f/D$$

where F is the focal ratio (f-number). The noise has contributions from **photon shot noise** ($\sqrt{N_e}$), **dark current**, **read noise** and **quantisation**, so

$$\mathrm{SNR} = \frac{N_e}{\sqrt{N_e + N_{dark} + \sigma_{read}^2 + \sigma_{quant}^2}}$$

For a pushbroom imager, the integration time is limited by the time the ground image takes to move one pixel: $t_{int} = \mathrm{GSD}/v_{ground}$. This is about 1 ms for a 7 m GSD at a ground speed of 7 km/s, and only about 50 µs for a 30 cm GSD. This is why very-high-resolution imagers need large apertures, **time-delay integration** with dozens of stages, or satellite "slow-down" manoeuvres that rotate the spacecraft to slow the apparent ground motion.

> **WORKED EXAMPLE — SNR OF A 10 M MULTISPECTRAL BAND**
>
> Consider a red band (centre 665 nm, $\Delta\lambda$ = 30 nm) viewing vegetation of radiance L = 50 W m^{-2} sr^{-1} µm^{-1} from 786 km with a 15 cm aperture, f/6.5 telescope (τ = 0.8), 7.5 µm pixels (η = 0.8), and t_{int} = 1.5 ms (10 m pixel at 6.7 km/s ground speed).
>
> $N_e = (\pi/4) \times 50 \times 0.8 \times 0.8 \times 0.030 \times (7.5\times10^{-6})^2 \times 1.5\times10^{-3} / 6.5^2 \times (0.665\times10^{-6}/(6.63\times10^{-34}\times 3\times10^{8}))$
>
> $= 0.785 \times 50 \times 0.64 \times 0.030 \times 5.6\times10^{-11} \times 1.5\times10^{-3} / 42.3 \times 3.34\times10^{18} \approx 2.6\times10^{5}$ electrons.
>
> With shot noise dominating, SNR $\approx \sqrt{2.6\times10^{5}} \approx 500$ (before other noise sources), comfortably above typical requirements of 100–200. Real designs check SNR at a dark reference radiance (e.g., over water) where the margin is much smaller — the reason ocean-colour sensors use large apertures and wider bands.

12.9.2 Pointing, Jitter and Microvibration

A high-resolution imager magnifies every tiny motion of the spacecraft. If the line of sight moves by more than a fraction of a pixel during integration, the image blurs. For a 30 cm GSD at 600 km, one pixel subtends 0.5 µrad (0.1 arcsec) — so **jitter** during the 50–100 µs line time must be held to a fraction of that. Sources include reaction-wheel imbalance (at the wheel spin frequency and harmonics), cryocooler compressors, solar-array drive steps and thermal "snaps". Mitigations include isolating wheels on vibration-damping mounts, balancing wheels to very tight tolerances, using control-moment gyroscopes with high rotor speeds above structural modes, quiet (counter-balanced) cryocoolers, **fine-steering mirrors** in the optical path, and line-of-sight stabilisation. JWST and Hubble use dedicated fine guidance sensors; high-resolution EO satellites use extremely stiff, thermally stable optical benches made of silicon carbide, Zerodur or carbon-fibre composites.

12.9.3 Field-of-View and Keep-Out Analysis

Every instrument has a field of view that must not be blocked by solar arrays, antennas or thruster plumes. It also has keep-out zones that must avoid the Sun (to prevent detector damage or stray light), the Earth limb or the Moon. Accommodation studies use 3-D models to compute fields of view, stray-light paths (with tools

such as FRED or ASAP), thruster-plume impingement, and contamination deposition. Thrusters are placed so that plumes do not reach cold optics, and sensitive instruments carry covers that open only after outgassing has subsided in orbit.

12.10 Instrument Case Studies

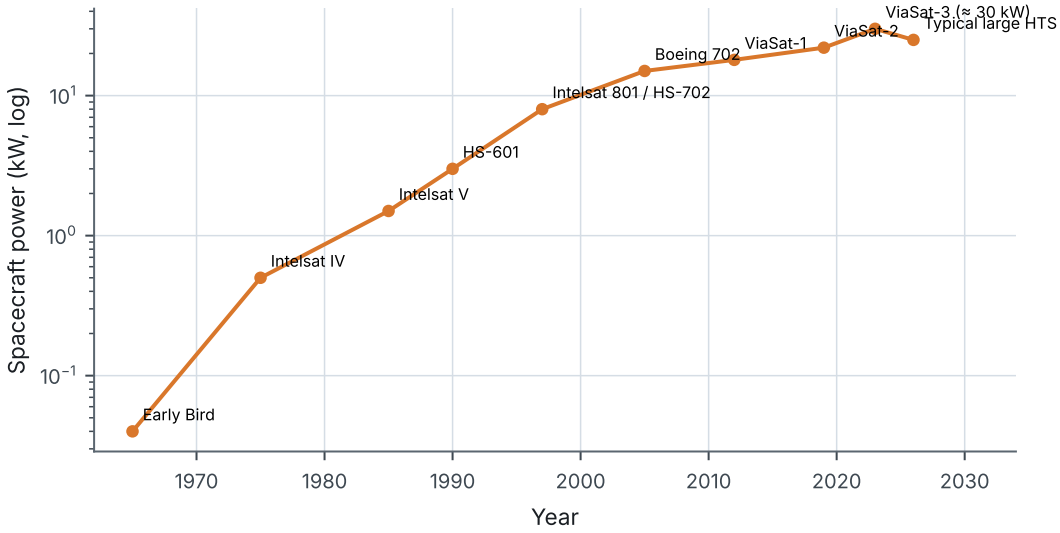
TABLE 12.2 Representative modern instruments and their key parameters.

Instrument (mission, launch)	Type	Key parameters	What it measures
OLI-2 (Landsat 9, 2021)	Pushbroom multispectral imager, 4-mirror anastigmat	9 bands 0.43–2.29 µm; 15 m pan, 30 m multispectral; 185 km swath; 14-bit	Land cover, agriculture, water quality, long-term record since 1972
TIRS-2 (Landsat 9)	Pushbroom thermal imager, QWIP detectors at ≈ 40 K	2 bands 10.9 and 12.0 µm; 100 m	Land-surface temperature, evapotranspiration
MSI (Sentinel-2A/B/C, 2015–2024)	Pushbroom TMA imager	13 bands 0.44–2.19 µm; 10/20/60 m; 290 km swath	Vegetation, coastal and inland waters
ABI (GOES-R series, 2016–2024)	Scanning GEO imager	16 bands 0.47–13.3 µm; 0.5–2 km; full disk every 10 min, mesoscale every 30–60 s	Weather, fire, volcanic ash, cloud properties
VIIRS (Suomi NPP, NOAA-20/21)	Whiskbroom scanning radiometer	22 bands 0.41–12.5 µm; 375 m / 750 m; day–night band	Weather, fires, night lights, ocean colour, aerosols
IASI-NG (MetOp-SG A, 2025)	Fourier-transform IR sounder	3.62–15.5 µm, 16 921 channels, 0.25 cm^{-1}	Temperature, humidity and trace-gas profiles
ATMS (JPSS)	Microwave sounder	22 channels 23–183 GHz	All-weather temperature and humidity profiles
SMAP radiometer (2015)	L-band radiometer, 6 m rotating mesh antenna	1.41 GHz, ≈ 40 km, 1 000 km swath	Soil moisture, freeze/thaw
ATLAS (ICESat-2, 2018)	Photon-counting lidar	532 nm, 10 kHz, 6 beams, ≈ 0.7 m along-track sampling	Ice-sheet, sea-ice and vegetation height
C-SAR (Sentinel-1, 2014–2025)	C-band SAR, active phased array 12.3 × 0.8 m	5.405 GHz; 5 × 20 m (IW mode), 250 km swath	Deformation, floods, ice, ships
TROPOMI (Sentinel-5P, 2017)	UV–SWIR imaging spectrometer	270–2 385 nm; 5.5 × 3.5 km; 2 600 km swath	NO_2, O_3, SO_2, CO, CH_4, aerosols
MAG (Swarm, 2013)	Vector fluxgate + absolute scalar magnetometers on boom	0.1 nT resolution	Geomagnetic field and its variations

These instruments show recurring design patterns: pushbroom imagers with TMA optics for land imaging, scanning imagers in GEO for weather, and FTS and grating spectrometers for atmospheric composition. Others are large deployable antennas for long-wavelength microwave measurements, and booms for magnetic measurements.

12.11 Large Communications Platforms: Power as the Measure

For communications satellites, available power largely determines capacity. Over six decades GEO communications satellites grew from the 40 W *Early Bird* (1965) to platforms of 20–30 kW (Figure 12.11). This growth was enabled by multi-junction solar cells, lithium-ion batteries, deployable radiators with loop heat pipes, and electric propulsion that frees mass for payload. Very-high-throughput satellites such as **ViaSat-3** (each about 6.4 t and about 30 kW, deploying a very large reflector) push the limits of what a single platform can do. However, the deployment of ViaSat-3 F1's reflector partially failed in 2023, showing that large deployables remain a risk.



Approximate end-of-life power of representative geostationary communications satellites.

FIGURE 12.11 Approximate power of representative geostationary communications satellites since 1965.

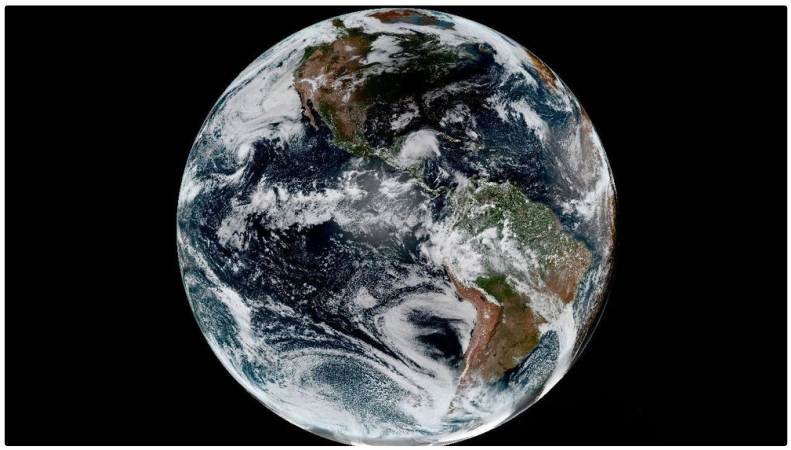


FIGURE 12.12 A GOES-R series full-disk image of the Western Hemisphere from the Advanced Baseline Imager in geostationary orbit — the kind of product made possible by a capable payload carried on a stable bus. Image: NASA/GSFC.

KEY TAKEAWAYS

- Most satellites use standard buses from product lines; trends are productisation, software-defined platforms and vertical integration; hosted payloads share buses to reduce cost.
- Payload–bus interfaces (mechanical, power, thermal, data, time, pointing, contamination/EMC) must be specified and verified carefully.
- Payloads are passive or active remote-sensing instruments, in-situ field and particle instruments, or service payloads; each spectral region reveals different physics, and atmospheric windows and absorption bands are both useful.
- Optical systems combine telescopes (refractors, Ritchey–Chrétien, TMA, segmented mirrors) with detectors chosen for wavelength (Si, InGaAs, InSb, HgCdTe, microbolometers) and cooled as needed; pushbroom imaging dominates modern EO.
- Microwave radiometers, radars, GNSS radio occultation and lidars measure the atmosphere and surface in all weather; magnetometers and particle instruments sample space plasmas.
- Performance is described by GSD, MTF, SNR, NEΔT, spectral resolution, radiometric accuracy and geolocation; calibration from the laboratory to the Moon and vicarious sites makes measurements trustworthy.

REVIEW QUESTIONS

1. Why have most satellite manufacturers moved to standard bus product lines? What are hosted payloads?
2. List the interfaces between a payload and its bus that must be specified and verified.

3. Which detector materials are used for visible, short-wave infrared and thermal-infrared imaging, and which require cooling?
4. Compare framing, whisk-broom and push-broom imaging architectures.
5. A push-broom imager at 500 km altitude has 8 μm pixels and a 4 m focal length. What is its ground sample distance at nadir? What aperture is needed for its diffraction-limited resolution at 0.6 μm to match that GSD?
6. How does GNSS radio occultation measure atmospheric temperature and humidity profiles?
7. Explain why calibration is needed before launch, in orbit and with vicarious sites, and give an example of each.
8. Why is a communication satellite's payload capability usually described by its power?